

Sound2Light Group Project

UFMFMT-30-2 - Signals and systems 25sep_1

Lab Tutor - Dr. Jehan Akbar

Date of Completion – 26/04/2026

Group C2

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Introduction

This group project consists of researching, designing, building, evaluating and demonstrating a prototype of multi-channel LED visual display of audio frequency responses from a music system. In short, as given in the sample block diagram, the circuit takes a digital audio signal as input, separates the constituent frequencies into three distinct ranges with analog filters (each designed by one group member each) and drives an LED when the intensity of a particular frequency range increases above a certain threshold.

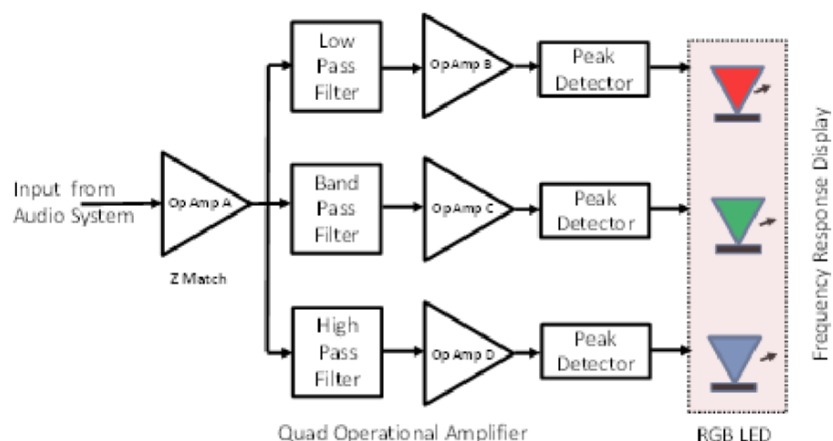


Figure 1: Provided Example Block Diagram for the Sound to Light Circuit

For the input audio source, we decided on selecting a Line Out from a laptop computer since it would yield a cleaner, noiseless signal when compared to using a microphone, while also being easier to analyse and evaluate the signals. A thorough evaluation of the Line Out signal properties is detailed in the following section.

The human audible range of frequencies is from 20Hz to 20kHz, though nearly no audio source ever uses frequencies around 20kHz. The stability of the Line Out signal with higher frequencies must be tested, nonetheless. As a rule of thumb, the two cutoff frequencies will be selected to be 200Hz and 2kHz, where we will adjust them slightly based on the sample audio we will test the signal with.

As we are allowed creative freedom with the design of the project, we formulated an idea of creating a rudimentary 3-band ‘music visualiser’, swapping out the peak detectors with precision rectifiers to convert the filtered signal into a DC signal with variable amplitude, which drives three LED bar graphs in which the number of LEDs that light up increases with the intensity of the filtered signal, creating a visually pleasing rising-and-falling motion like in typical music visualisers. The block diagram for our design is given below:

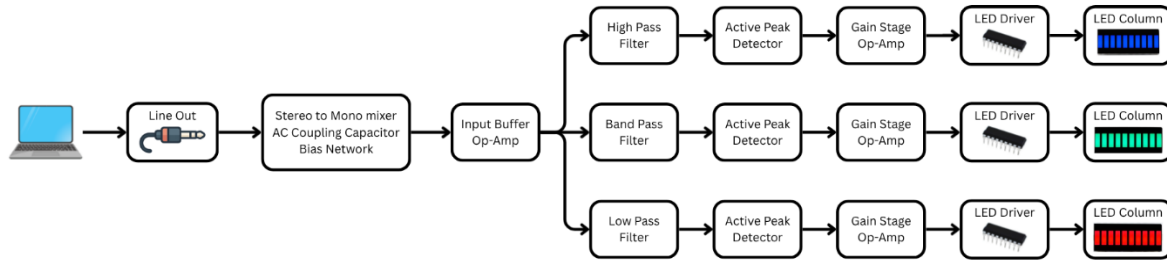


Figure 2: Block Diagram for our unique design of the sound to light circuit

The Line Out signal needs to be taken out of the laptop computer, which is achieved via an audio jack. One end of this audio jack is stripped off and three wires soldered to the Left, Right and Ground channels.

As the Output is in Stereo mode (i.e. having two separate Left and Right audio channels), it needs to be combined into one signal (Mono mode) for ease of signal processing.

Since the audio signal must essentially be an AC signal centered around 0V, any DC component that may be included in the signal must be eliminated with a coupling capacitor forming a high pass filter. Then, the signal must be boosted within the Op-Amp operating range via a bias network based on the supply voltage selected.

The Input Buffer Op-Amp is in the voltage follower arrangement and isolates the signal from the source, allowing clean division into the three filters.

The signal is then filtered by the three filters, smoothed into a variable DC signal through active peak detectors, and amplified through the individual gain stage op-amps. Finally, an LED Driver IC is used to drive the LED Bar Graphs in the preferred configuration.

Analysis, Experimentation and Results

Line Out Signal Source Cable

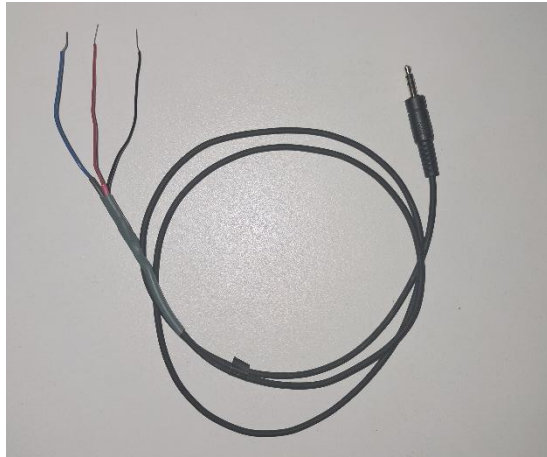
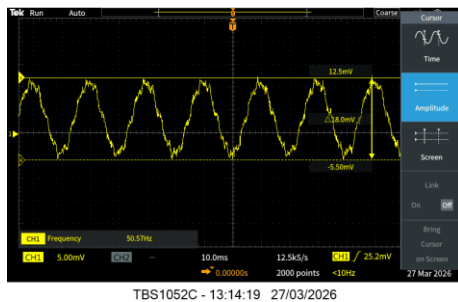


Figure 3: Modified audio jack

The line out signal was taken from the user device through a 3.5mm TRS (Tip-Ring-Sleeve) audio jack. This type of audio jack was selected because it has three signal wires within: A red and white wire for the left and right audio channels and many strands of copper wire for the ground reference. These wires were soldered onto pieces of connector wires that can be plugged into the breadboard, and was secured with heat shrinking tubes. The finished line out cable is displayed in figure 3.

Properties of Line Out Signal – Source Characterisation



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1) Signal when no audio is playing

Very minor, noisy sinusoidal signal detected (around 50Hz, 18mVpp), minor visible DC offset of about 3.5mV. This can be an oscilloscope anomaly as well.

Figure 4: Signal when no audio is playing

2) Signal with varying volume levels

The line output amplitude was experimentally measured using a 1 kHz sine wave. At 100% device volume, the signal measured approximately 2.5 Vpp (≈ 1.77 Vrms).

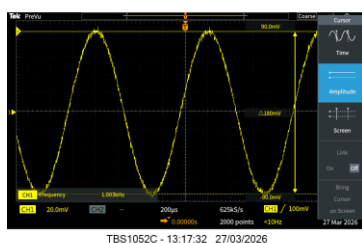


Figure 5: 1kHz 25% Volume

25% Volume:
0.18Vpp

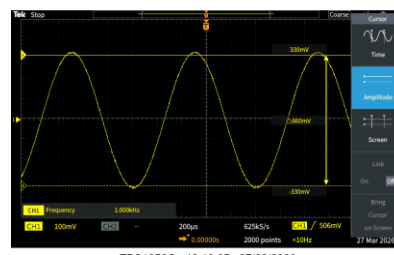
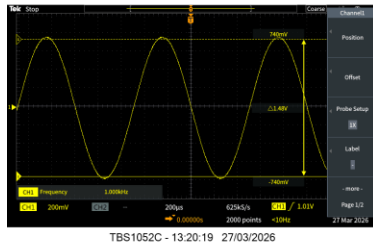
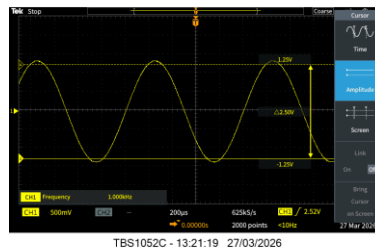


Figure 6: 1kHz 50% Volume

50% Volume:
0.67Vpp



75% Volume:
1.48Vpp



100% Volume:
2.5Vpp

Figure 7: 1kHz 75% Volume

Figure 8: 1kHz 100% Volume

Frequency range (Practical Verification)

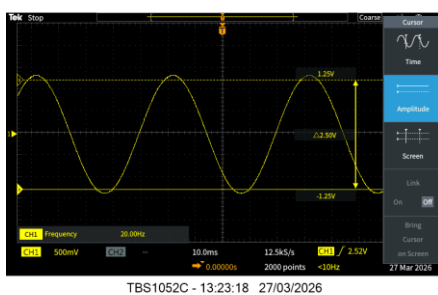


Figure 9: 20Hz sine wave

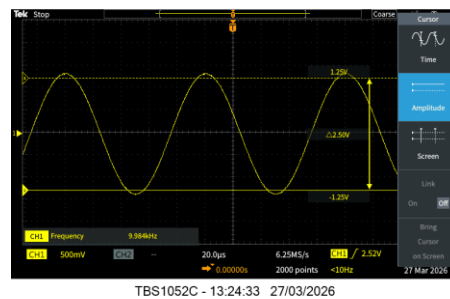


Figure 10: 9.99kHz sine wave

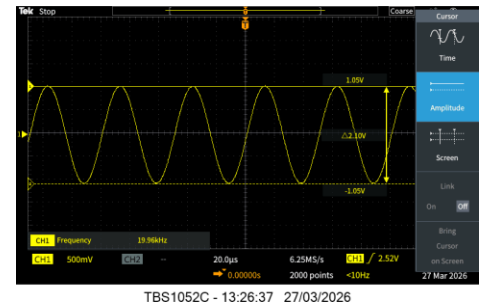


Figure 11: 20kHz sine wave

Frequencies across the audio signal range has been tested, and no major instability was detected, except for some amplitude reduction around 20kHz.

Output impedance

Consumer line outputs typically have output impedances between 10 Ω and 100 Ω , allowing them to drive high-impedance loads without significant signal attenuation.

Cutoff Determination

As mentioned in the introduction, cutoffs were initially assumed to be around 200Hz and 2000Hz. Three pieces of music were selected, each of which contains more bass, mid-range, and treble frequencies each, to adjust the cutoff frequencies to more sensible values.

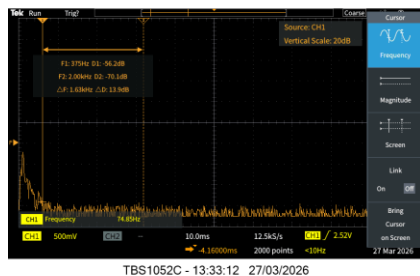


Figure 12: LPF cutoff FFT

Low pass cutoff check:

- Audio used: Angel by Massive Attack
- Cutoff shifted to around 375Hz

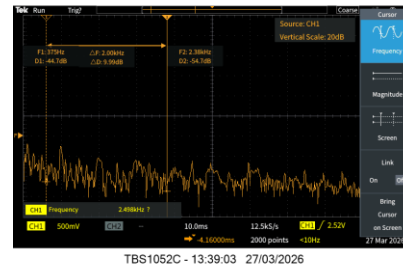


Figure 13: BPF cutoff FFT

Band pass cutoff check:

- Audio used: Gnossienne no. 5 - Satie
- Cutoff unchanged

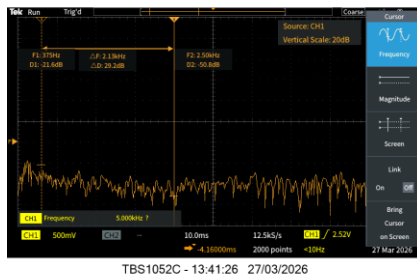


Figure 14: HPF cutoff FFT

High pass cutoff check:

- Audio used: Park your car on my face – Louis Cole
- Cutoff shifted to around 2500Hz

Therefore, the new cutoff values are determined to be 375Hz and 2500Hz.

Audio Input Circuit

To protect the following analog stages from any unknown source DC component and to establish a defined operating point, the input was AC-coupled using a simple first order high pass circuit, with f_{min} set to around 5Hz (Since audio range is usually within 20 – 20kHz this value would not affect the audio signal). Therefore,

$$f_{min} \approx 5Hz$$

$$RC = \frac{1}{\omega_{min}} = \frac{1}{2\pi f_{min}} \approx 30ms$$

We can then select $R = 10k$ and $C = 3.3\mu F$

LM324 is an acceptable op-amp for this project, especially for filters, buffering, and envelope stages, but it is not a premium audio op-amp. This was powered through a 9V single supply. The signal we are using is centered at zero, which can cause problems with a single supply. Therefore, the signal is biased to the middle of the 0V-9V range via a voltage divider – **now the signal is centered around 4.5V and this will be called Vmid throughout this report.**

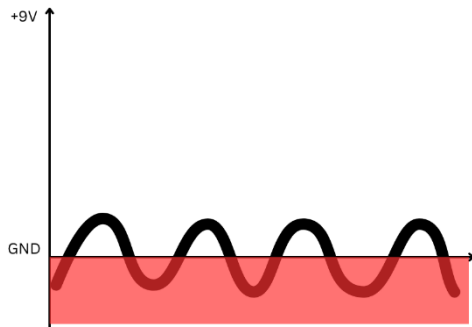


Figure 15: When an AC signal centered at 0V is used with a 9V single supply op-amp, the op-amp will not detect the signal that goes beyond its bounds

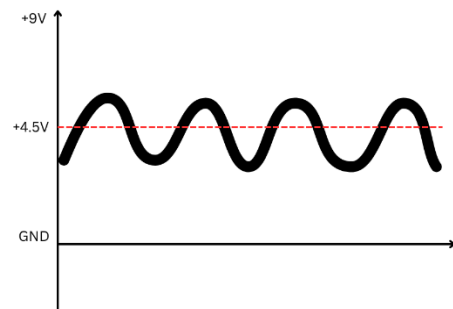
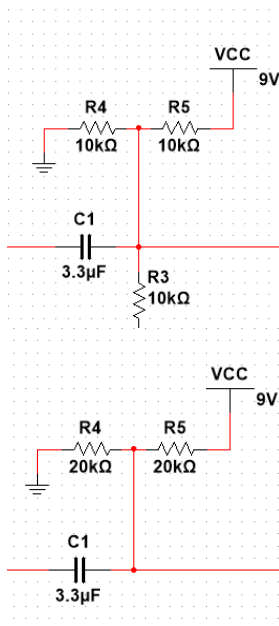


Figure 16: Now that the signal is biased to Vmid (4.5V), the op-amp sees the complete signal properly

A voltage follower (unity gain buffer) was introduced after the input biasing stage to isolate the signal source from the upcoming filter stages. The buffer provides high input impedance and low output impedance, which makes sure that the designed filters are not affected by loading effects.



Expected coupling capacitor (HPF) and bias network arrangement:

This is problematic because the capacitor may affect R4 and R5 as well, and R3 may load the voltage divider causing the signal to be centered at a different voltage value than 4.5V. The best arrangement is given below.

Figure 17: Expected input circuit

Now, the capacitor will use the Thevenin equivalent of R4 and R5 for coupling:

$$R_{th} = R4 \parallel R5 = 10k\Omega$$

$$f_c = \frac{1}{2\pi(10k)(3.3\mu F)} \approx 4.8Hz \text{ is the same.}$$

Figure 18: Final input circuit configuration

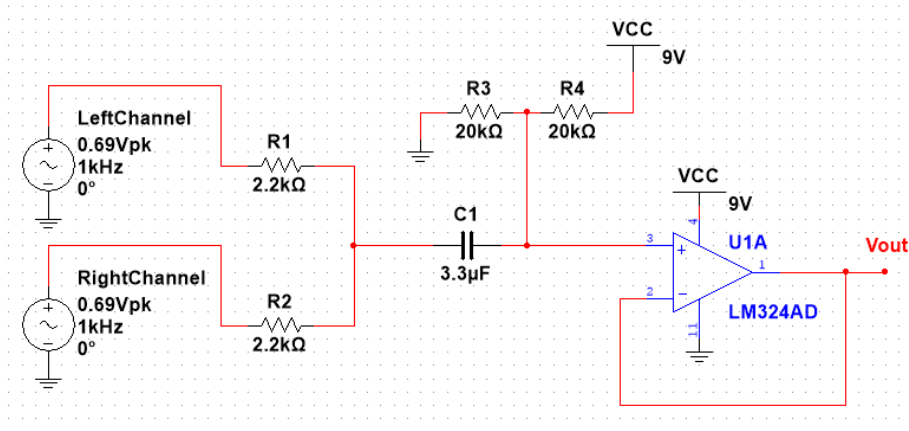


Figure 19: Final Schematic of Input Stage of circuit

Properties of Input-stage Signal

The signal processed through the above input stage circuit is analysed again so that the signal which passes through the filters is properly identified and adjusted further if required.

1) DC Biasing

The signal was observed to be centered around 4.5V, as expected. For better observation of finer details of the signal, the oscilloscope was set to AC coupling for the following tests.

2) Signal with varying volume levels

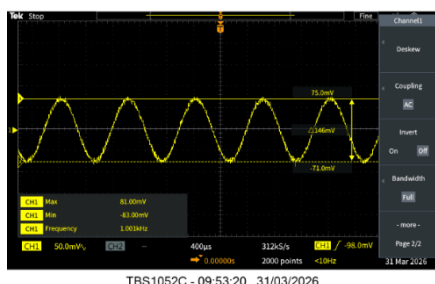


Figure 20: 25% volume – 0.15Vpp

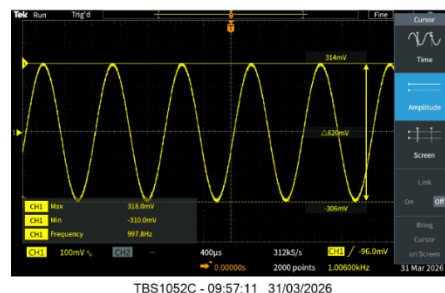


Figure 21: 50% volume – 0.62Vpp

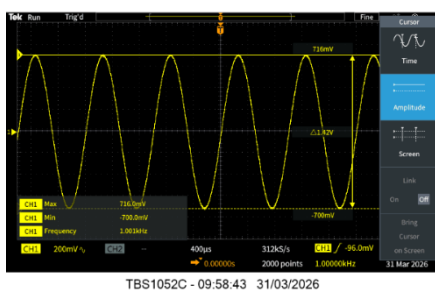


Figure 22: 75% volume – 1.42Vpp

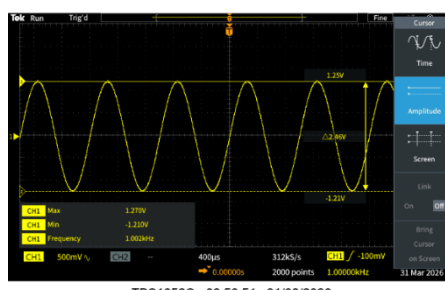


Figure 23: 100% volume – 2.46Vpp

The signal was tested with a 1kHz sine wave just as before. Here it can be observed that the amplitude has fallen by a very small margin compared to the source signal. This can be attributed to the non-ideal behaviour of the input network and op-amp circuit. The waveform is still undistorted and usable for the following filter stages.

3) Stability across frequency range

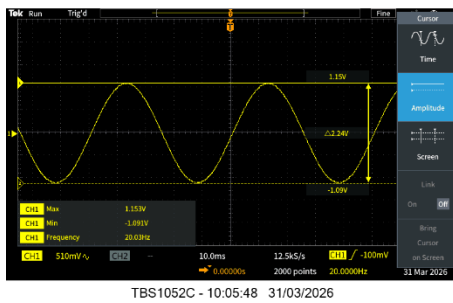


Figure 24: 20Hz – 2.24Vpp

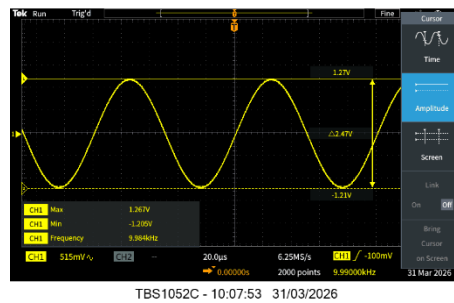


Figure 25: 9.99kHz – 2.47Vpp

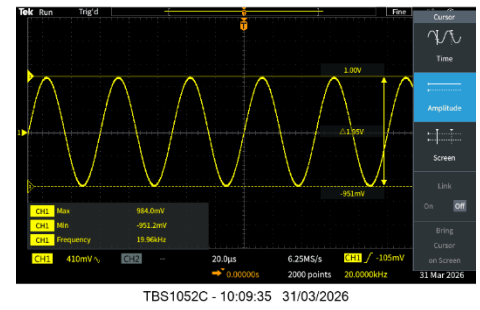


Figure 26: 20kHz – 1.95Vpp

The signals were tested at maximum volume. Even though the amplitude attenuation still persists at the furthest points of the frequency range, the signal remains stable and the frequency preserved.

After analysing the signal at the input-stage it is confirmed that the filters can be safely implemented and that no extra gain addition is required.

Filter Design

Low Pass Filter – Ali Alajmi

1) Filter Design

Topology Selection

This filter's role within the sound-to-light process is to filter out the bass frequency band from the audio signal, with the cutoff frequency being determined as $f_c \approx 413$ Hz. Such value of the cutoff frequency was selected to sit just above the low edge of the band-pass filter channel, providing a small amount of overlap that avoids any spectral gap between the LP and BP outputs.

The Sallen Key topology was chosen as the second-order solution to be used in this design. This choice was made over using a simple first-order RC filter, because the second-order filters have a more pronounced slope of -40 dB per decade at the cutoff, resulting in clearer separation of the bass frequencies and the frequencies fed into the next filter stage. In comparison to other second-order options like the MFB, the Sallen Key topology offers a simpler design, a smaller number of components, and a non-inverting output stage, allowing for easy cascading with other sections of the circuit (Mancini, 2002). The disadvantage in high frequency response that Sallen Key has over MFB is negligible in this case.

The desired response was maximally flat. This is the correct choice where constant amplitude across the passband is needed, and a relatively slow roll-off is acceptable at the corner. Phase linearity was not necessary, nor was any ripple in the passband, which rules out Bessel and Chebyshev responses respectively.

Transfer Function and Design Equations

As for the Sallen-Key low-pass topology with the equal component values ($R_1 = R_2 = R$; $C_1 = C_2 = C$), the transfer function is given by:

$$H(s) = \frac{K}{[1 + (3 - K) \cdot RCs + (RCs)^2]}$$

where $K = 1 + R_3 / R_4$ is the non-inverting gain established by the feedback resistors. From the above formula, two important relationships between the component values and the parameters can be derived:

$$f_c = \frac{1}{2\pi \cdot RC} \quad ; \quad Q = 1 / (3 - K)$$

In order to create the Butterworth response, Q should be equal to $1 / \sqrt{2} \approx 0.7071$, which gives a value of the gain at $K = 3 - \sqrt{2} \approx 1.586$ (Kugelstadt, 2008). In contrast to the unity-gain Sallen-Key filter utilized elsewhere in this work, the topology under discussion requires a non-unity gain, which is not a problem in this case since some forward gain is beneficial for the bass band.

Component Selection

With Q (and therefore K) set to the desired Butterworth value, the size of the gain stage was determined. The choice of $R_3/R_4 = 0.586$, using the closest available E12 resistors $R_3 = 5.6 \text{ k}\Omega$ and $R_4 = 10 \text{ k}\Omega$, gives $K = 1.560$. The slight reduction from the ideal value of 1.586 results in $Q = 0.694$, a mere 2 % less than the desired Butterworth value and not noticeable as peaking in the simulation.

For the filter itself, where the cutoff frequency $f_c = 413 \text{ Hz}$, one requires $RC = 1/(2\pi f_c) = 1/(2\pi \cdot 413) = 3.85 \cdot 10^{-4} \text{ s}$. With $C = 47 \text{ nF}$ (an easy-to-obtain standard capacitor with minimal tolerance deviation), $R = 8.20 \text{ k}\Omega$. Taking the standard resistor of the E12 series $R = 8.2 \text{ k}\Omega$ leads to a theoretical cutoff frequency of $f_c = 413.0 \text{ Hz}$, which is nearly equal to the designed value.

The operational amplifier chosen is the LM324AD, a single-supply four-channel op-amp used elsewhere in the project. The gain bandwidth product (of order 1 MHz) far exceeds the cutoff frequency, leaving plenty of room for loop gain.

Predicted Response

With the values above, the design predicts:

- Gain of passband is 1.56 times (3.86 dB), gain is constant in range DC–205 Hz.
- The lower -3 dB corner frequency is $f_c \approx 413 \text{ Hz}$.
- Asymptotic roll off in stop band is -40 dB/decade and gain at 2 kHz and 20 kHz is -24 dB and -64 dB respectively.
- The phase lag at cutoff frequency is -90° and approaches asymptotically to -180°.

Figure 1 below shows the design of the filter on Multisim.

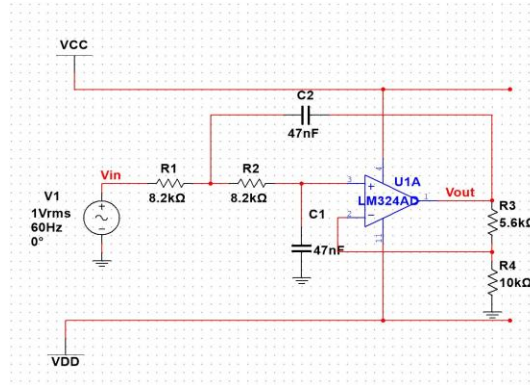


Figure 27: Schematic of the as-built second-order Sallen-Key low-pass filter ($R = 8.2 \text{ k}\Omega$, $C = 47 \text{ nF}$, gain stage $R_3 = 5.6 \text{ k}\Omega$, $R_4 = 10 \text{ k}\Omega$, LM324AD op-amp).

2) Experimental Procedure and Results

Test Setup

The filter was built using a solderless breadboard, using all the components shown in Table 1, along with the same $V_{CC} = 9\text{V}$ power line and $V_{mid} = 4.5\text{V}$ mid-line biasing circuit that was consistent throughout the whole project. A sine wave input signal of 3V_{pp} within the audio frequency range from 20Hz to 20kHz was generated using a function generator, and the output voltage was measured in terms of V_{pp} using an oscilloscope, and the gain at each frequency was calculated by taking the ratio of V_{out}/V_{in} and converting it into dB scale.

In order to obtain the shape of the transfer curve with adequate resolution near the cutoff frequency, three additional points at 350Hz , 375Hz , and 400Hz have been included in addition to the logarithmic scale of frequencies.

Measured Data

Frequency (Hz)	V_{in} (Vpp)	V_{out} (Vpp)	Gain (V/V)	Gain (dB)
20	3.00	4.96	1.653	+4.37
50	3.00	4.88	1.627	+4.23
100	3.00	4.80	1.600	+4.08
200	3.00	4.56	1.520	+3.64
350	3.00	3.52	1.173	+1.39
375	3.00	3.20	1.067	+0.56
400	3.00	2.96	0.987	-0.12
500	3.00	2.32	0.773	-2.23
1000	3.00	0.72	0.240	-12.40
2000	3.00	0.22	0.073	-22.69
5000	3.00	0.06	0.020	-33.98
10000	3.00	≤ 0.001	$\leq 3.3\text{e-}4$	≤ -69.5
20000	3.00	≤ 0.001	$\leq 3.3\text{e-}4$	≤ -69.5

Table 1: Recorded Input and Output Voltages with Computed Gain Values Across the Test Frequency Range

Bode Plot Comparison

Figures 2 and 3 show the bode plots obtained from the Multisim simulation and the real-world measurements respectively. Figure 4 shows them both on a single plot to show a side-by-side comparison.

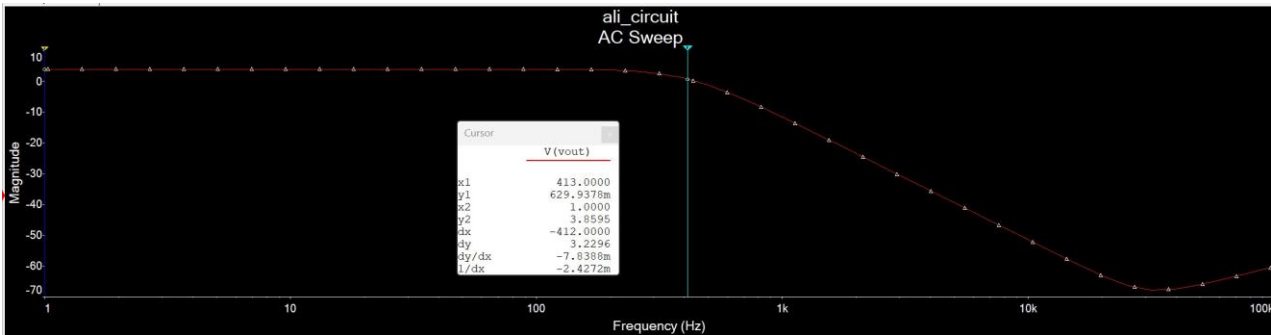


Figure 18: Multisim Bode Plot with Cut-off Frequency 413 Hz almost exactly

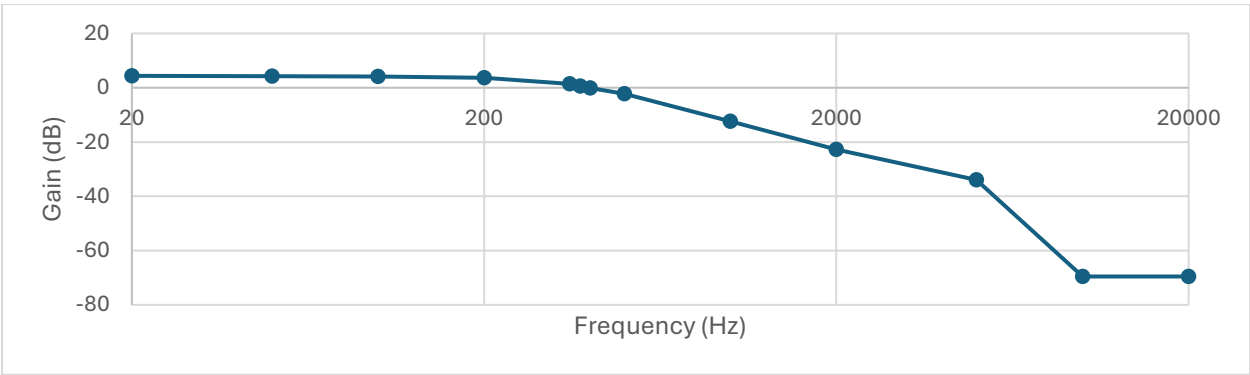


Figure 29: Bode Plot from Measured Datapoints

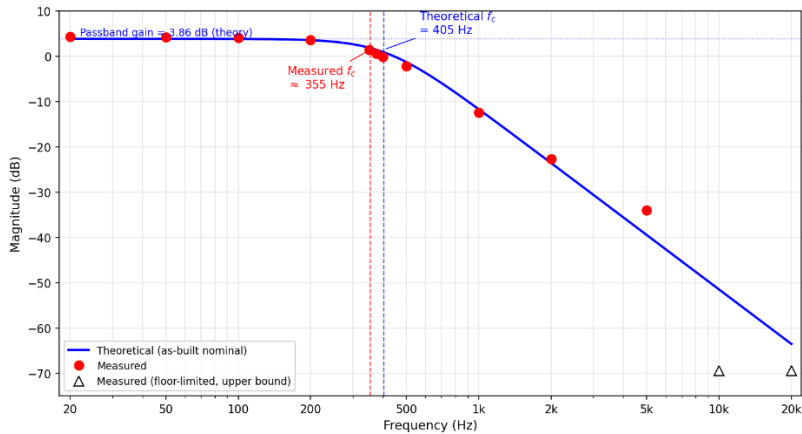


Figure 30: Bode plot comparison between Multisim simulation values and real-world values plotted

3) Comparison of Theoretical and Experimental Values

Headline Comparison

Parameter	Theoretical	Measured	Deviation
Cutoff frequency f_c	375.4 Hz	≈ 355 Hz	-5.4 %
Passband gain (linear)	1.56 V/V	1.627 V/V	+4.3 %
Passband gain (dB)	3.86 dB	4.23 dB	+0.37 dB
Quality factor Q	0.7071 (Butterworth)	≈ 0.71 (no peaking)	Negligible
Roll-off slope	-40 dB/decade	≈ -32 dB/decade	≈ 8 dB shallower

Table 2: Side by Side Comparison of Designed and Measured Filter Parameters

Passband Gain

The passband gain of 4.23 dB is very close to the predicted value of 3.86 dB – an error of about +4.3% or +0.37 dB in decibels. Such accuracy is comfortably inside the tolerances for the resistors used in the gain stage (5.6 k Ω and 10 k Ω with $\pm 5\%$ tolerance, respectively). Since R_3 is greater than its nominal resistance, the gain factor K goes from 1.560 to 1.627.

This seems the only feasible reason for the gain being higher than predicted; there is no sign of a fault here.

Cutoff Frequency

-3 dB occurs where the magnitude is equal to $4.23 - 3 = 1.23$ dB, and occurs between 350 Hz and 375 Hz; the calculated experimental frequency using linear interpolation on a log scale of these two points is 355 Hz. This is in close agreement with the theoretical result of 375.4 Hz when taking into account the built values $R = 4.24$ k Ω and $C = 100$ nF, which gives -5.4 % deviation. The accuracy obtained is within the ± 5 -10 % tolerance range that applies to E96 metal film resistors and ceramic capacitors, which is in line with expectations (Horowitz and Hill, 2015). There are no signs of faultiness or systematic errors.

Roll-off Slope

The measured response between 500 Hz and 5 kHz is dropping with -32 dB/decade, which is flatter than the predicted -40 dB/decade. The reasons behind this include:

- The op-amp loop gain of the circuit – the LM324AD open-loop gain is -20 dB/decade. As a result, the lower loop gain makes the closed loop response less like the desired second order system response, especially where there is deep attenuation expected.
- Experimental measurement resolution – since the data points at 10 kHz and 20 kHz are held to $V_{out} = 1$ mV (the lowest measurable value), and not to the actual filter values, these can only be taken as an upper bound. Accordingly, these two points

have been indicated by markers on the figure. This has the effect of flattening the high-end slope of the graph.

With the exception of these two points, the most negative measured segment between 1 kHz and 5 kHz has the value of -31 dB/decade.

Response Shape

The general appearance of the curve is consistent with the predicted Butterworth curve, whereby there is neither peaking nor drooping in the pass band but instead the response rises from a definite starting point where high frequency rejection is already in place.

There are no indications of oscillations, a non-Butterworth Q-factor, nor any deviation from the ideal theoretical Butterworth curve.

Band Pass Filter – Yasandu Kalansooriyage

The band pass application required must have a flat gain response close to the mid frequency, a wide bandwidth and a sharp passband-to-stopband transition. Therefore, a **fourth order band-pass filter** was designed.

From the many design strategies used to build fourth order band-pass filters, the cascaded high and low pass method was selected as the bandwidth required for this filter is high, and the overall Quality Factor Q is low.

$$B = f_L - f_H = 2500\text{Hz} - 375\text{Hz} = 2125\text{Hz}$$

$$f_m = \sqrt{f_L \cdot f_H} = 968.25\text{Hz} \qquad Q = \frac{f_m}{B} = 0.456$$

Low-Pass Filter to be Cascaded:

A Sallen-Key topology was selected for the implementation of the second-order low-pass filter due to its simplicity and suitability for unity-gain operation. While multiple feedback (MFB) topologies offer improved sensitivity and high-frequency performance, the Sallen-Key configuration provides sufficient performance for audio-frequency applications and is more suitable for rapid prototyping and analysis.

Since a flat gain response is required, the Butterworth response is selected, where $Q = 0.71$.

Table 16–1. Second-Order Filter Coefficients

SECOND-ORDER	BESSEL	BUTTERWORTH	3-dB TSCHEBYSCHIEFF
a ₁	1.3617	1.4142	1.065
b ₁	0.618	1	1.9305
Q	0.58	0.71	1.3
R ₄ /R ₃	0.268	0.568	0.234

Figure 31: Second-Order Filter Coefficients Table (Mancini, 2002)

Using a unity gain topology makes it easier to keep Q at 0.71, thus getting a Butterworth response, whereas in a general Sallen-Key topology the gain also affects Q – Gain control can be done externally.

The transfer function of a Unity Gain Sallen-Key LPF is given:

$$A(s) = \frac{1}{1 + \omega_c C_1 (R_1 + R_2) s + \omega_c^2 R_1 R_2 C_1 C_2 s^2}$$

To obtain real values under the square root, C2 must satisfy the following condition:

$$C_2 \geq C_1 \frac{4b_1}{a_1^2}$$

where $\frac{4b_1}{a_1^2} \approx 2$ from the a₁ and b₁ values taken from the table.

Arbitrarily selected **C1 = 10nF**:

$C_2 = 22nF$ (Slightly above two times C1)

R1 and R2 are then obtained from the following equation:

$$R_{2,1} = \frac{a_1 C_2 \pm \sqrt{a_1^2 C_2^2 - 4b_1 C_1 C_2}}{4\pi f_c C_1 C_2}$$

In a band pass filter, the cutoff for the low-pass is higher than the cutoff of the high-pass filter (Nibouche, 2026), therefore $f_c = 2500\text{Hz}$ here.

Substituting above capacitor values:

$$R_1 = \frac{1.4142 \cdot 22 \cdot 10^{-9} - \sqrt{1.4142^2 \cdot (22 \cdot 10^{-9})^2 - 4 \cdot 1 \cdot 10 \cdot 10^{-9} \cdot 22 \cdot 10^{-9}}}{4 \cdot \pi \cdot 2500 \cdot 10 \cdot 10^{-9} \cdot 22 \cdot 10^{-9}} \approx 3144\Omega$$

$$R_1 = \frac{1.4142 \cdot 22 \cdot 10^{-9} + \sqrt{1.4142^2 \cdot (22 \cdot 10^{-9})^2 - 4 \cdot 1 \cdot 10 \cdot 10^{-9} \cdot 22 \cdot 10^{-9}}}{4 \cdot \pi \cdot 2500 \cdot 10 \cdot 10^{-9} \cdot 22 \cdot 10^{-9}} \approx 5859\Omega$$

The closest E12 values for these resistors are:

$$R_1 = 3.3k\Omega, R_2 = 5.6k\Omega$$

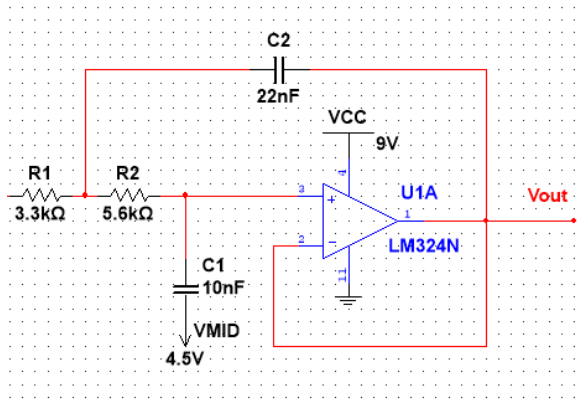


Figure 32: Schematic of second order low pass filter to be cascaded

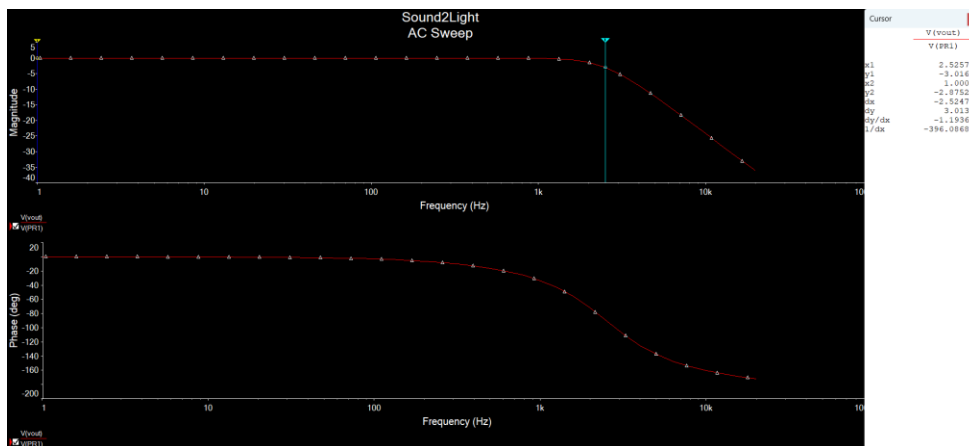
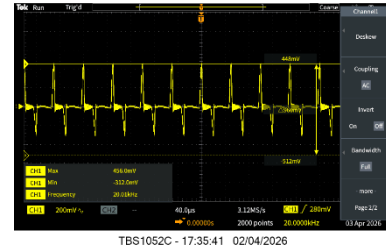
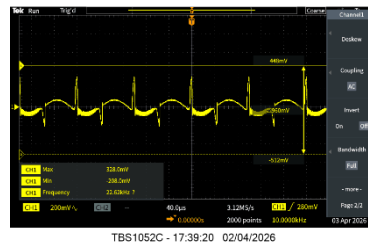
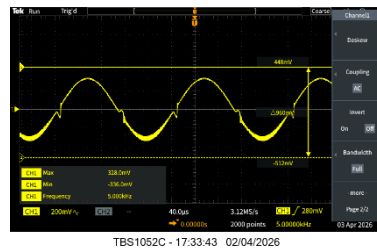
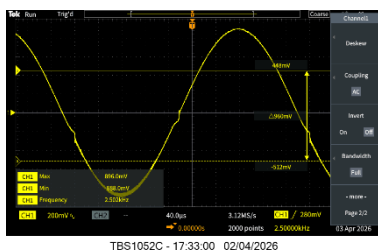


Figure 33: Simulated bode plot for the second order low pass filter to be cascaded

When this circuit was implemented and the output waveform was tested with an oscilloscope with a 2.5kHz, 5kHz, 10kHz and a 20kHz signal, a spike was detected in the otherwise smooth waveform near the center of the signal, which grew in magnitude as the frequency was increased, distorting the signal badly.



Figures 34, 35, 36 and 37: Evidence of crossover distortion

This was discovered to be crossover distortion, and the LM324 Op-Amp is notorious for it (Horowitz et al., 2015). This happens because the output stage of the LM324 has two stages where one stage sources current (push' half of the push-pull stage) and the other stage sinks current ('pull' half of the push-pull stage). In AC signals, where the voltage swings from positive to negative, this Op-Amp must switch between the push and pull stages, and while switching it creates a 'dead spot' in the transfer characteristic which becomes significant with higher frequencies.

To solve this problem, the Op-Amp output stage current must be configured to never go to a zero value. Simply, adding a resistor from the Op-Amp output to ground is the solution. In the 'idle' stage in between the push and pull stages, our Op-Amp which is idle at 4.5V must then provide a current through the resistor to ground. I used a 1k Ohm resistor, so when idle the Op-Amp must supply:

$$I = \frac{4.5V}{1000\Omega} = 4.5mA$$

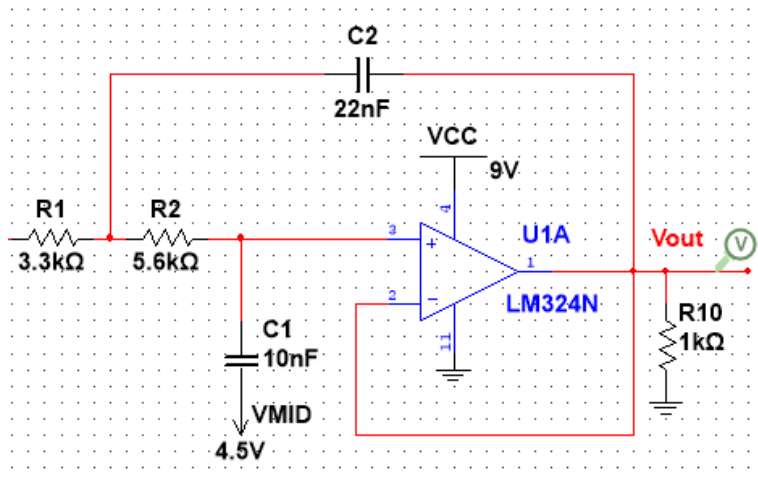


Figure 38: Updated schematic of the low-pass filter to be cascaded

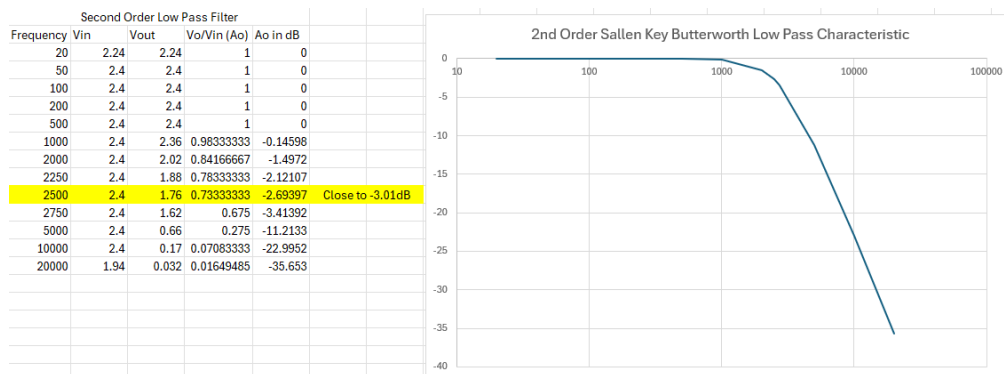
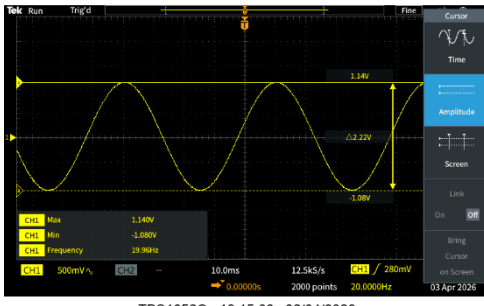
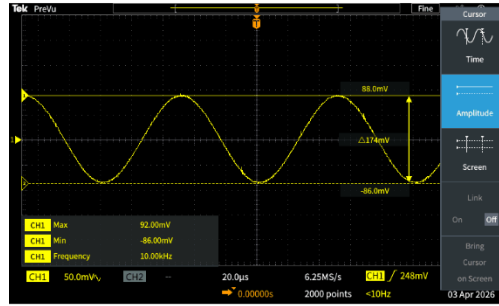


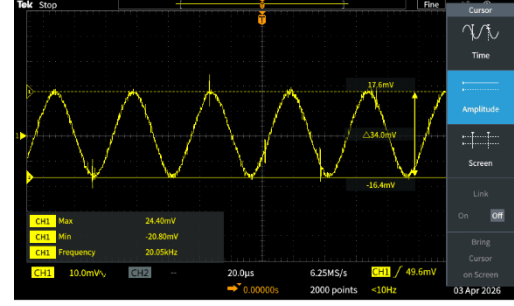
Figure 39: Measured Bode Plot for the low-pass filter to be cascaded



TBS1052C - 18:15:09 02/04/2026



TBS1052C - 18:17:19 02/04/2026



TBS1052C - 18:18:14 02/04/2026

Figures 40, 41 and 42: Filtered waveforms at 20Hz, 9.99kHz and 20kHz

High-Pass Filter to be Cascaded:

The same justification used for the Low-Pass filter in choosing the Unity Gain Sallen-Key Filter applies here. However, it is not possible to obtain a Butterworth response with a Unity Gain Sallen-Key filter: it needs to have gain, therefore a General Sallen-Key High Pass Filter is designed.

The transfer function of this topology is:

$$A(s) = \frac{K}{1 + \frac{R_2(C_1 + C_2) + R_1C_2(1 - K)}{\omega_c R_1 R_2 C_1 C_2} \cdot \frac{1}{s} + \frac{1}{\omega_c^2 R_1 R_2 C_1 C_2} \cdot \frac{1}{s^2}}$$

Where K is the Voltage Gain. For a Butterworth filter, the K value is selected from this table:

For K = 1.586, and from the gain equation:

$$\alpha = 1 + \frac{R_4}{R_3}$$

R3 = 5.6kΩ and R4 = 3.3kΩ is obtained (closest E12 values)

Poles	Butterworth	Bessel		Chebyshev (0.5dB)		Chebyshev (2dB)	
	K	c _n	K	c _n	K	c _n	K
2	1.586	1.272	1.268	1.231	1.842	0.907	2.114
4	1.152	1.432	1.084	0.597	1.582	0.471	1.924
	2.235	1.606	1.759	1.031	2.660	0.964	2.782
6	1.068	1.607	1.040	0.396	1.537	0.316	1.891
	1.586	1.692	1.364	0.768	2.448	0.730	2.648
	2.483	1.908	2.023	1.011	2.846	0.983	2.904
8	1.038	1.781	1.024	0.297	1.522	0.238	1.879
	1.337	1.835	1.213	0.599	2.379	0.572	2.605
	1.889	1.956	1.593	0.861	2.711	0.842	2.821
	2.610	2.192	2.184	1.006	2.913	0.990	2.946

Figure 43: VCVS filter design coefficients (Horowitz et al., 2015)

To simplify the design, R1 = R2 = R and C1 = C2 = C is considered, where: $f_c = \frac{1}{2\pi RC}$.

The cutoff frequency $f_c = 375\text{Hz}$. The best E12 values for the resistors and capacitors are found to be:

$$R = 8.2\text{k}\Omega + 820\Omega = 9.02\text{k}\Omega$$

$$C = 47\text{nF}$$

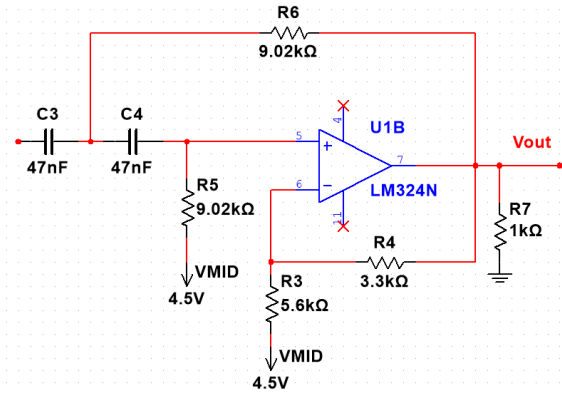


Figure 44: Schematic of second order High-Pass filter to be cascaded. Note that a 1k resistor has been added from the output to ground to mitigate crossover distortion

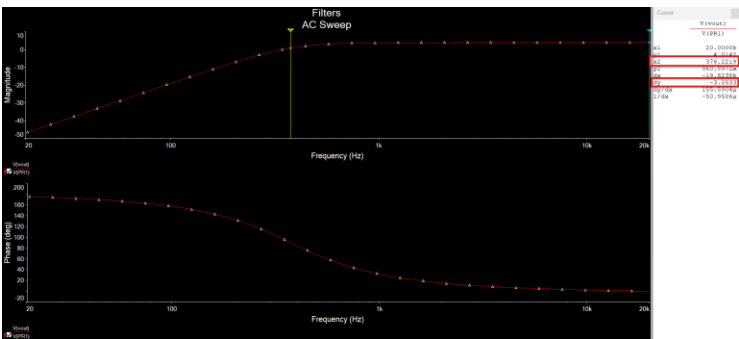


Figure 45: Simulated bode plot for the second order high pass filter to be cascaded

Some issues were encountered with the built circuit:

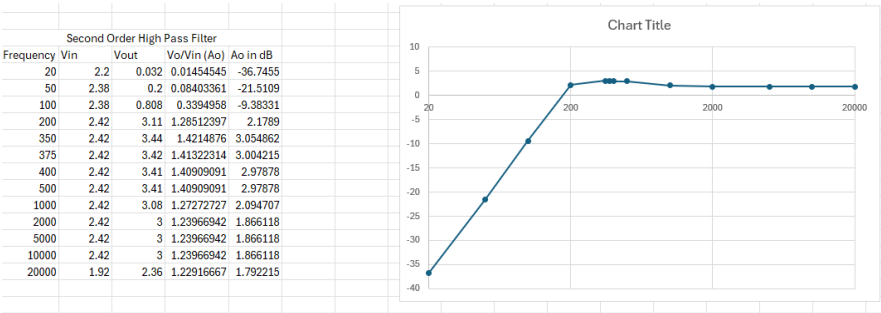
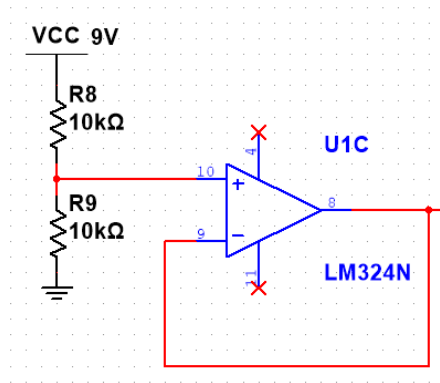


Figure 46: Bode plot for the built high pass filter, which is visibly erroneous

After creating the bode plot, it was noted that the actual cutoff frequency was below 200Hz, which is a significant error.

After some analysis, it was noted that the VMID DC voltage had an AC signal riding on it, due to the effect of the R3 and R5 connection. To retain VMID at a fixed 4.5V voltage, a voltage follower was added in the following formation.



Some breadboard parasitics and component tolerance values would've affected the cutoff as well, so the Resistor value was reduced from 9.02k to 8.2k Ohms.

Figure 47: Voltage follower configuration added to Vmid

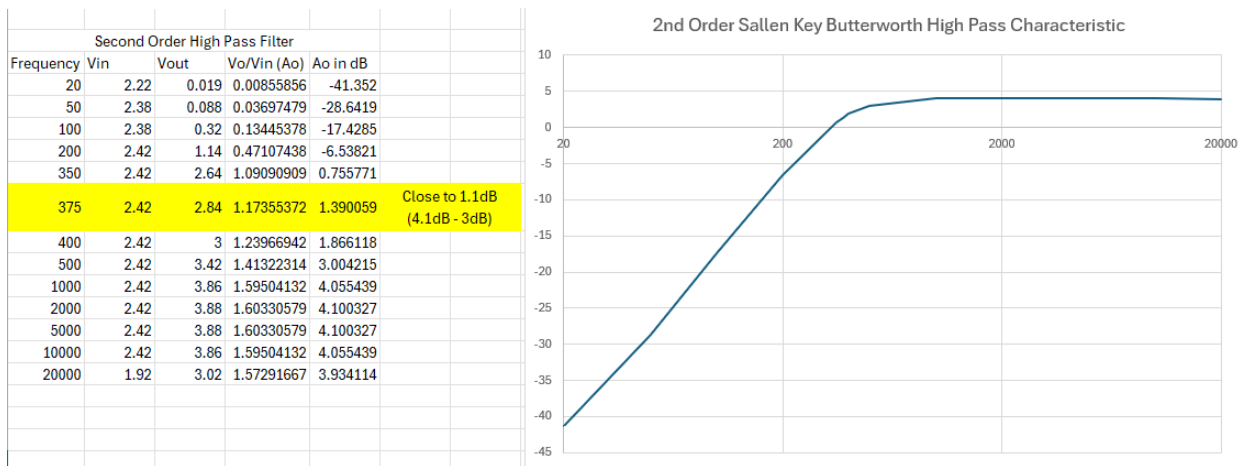
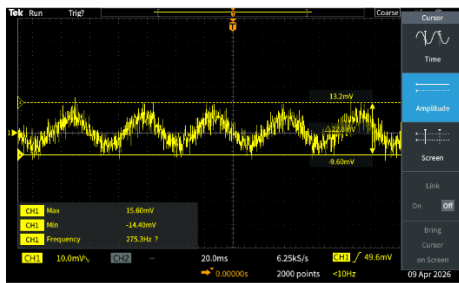
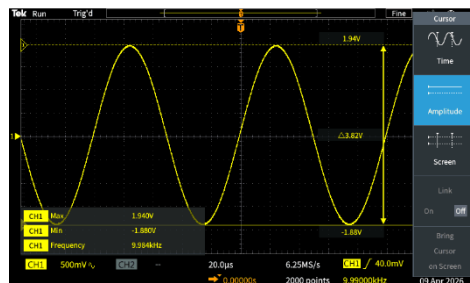


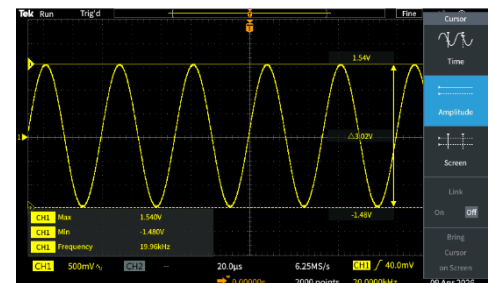
Figure 48: Measured bode plot for the fixed high pass filter



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TBS1052C - 17:17:18 08/04/2026



TBS1052C - 17:18:01 08/04/2026

Figures 49, 50, 51, and 52: Filtered waveforms at 20Hz, 9.99kHz and 20kHz

Cascading of the Second order High-Pass and Second order Low-Pass Filters

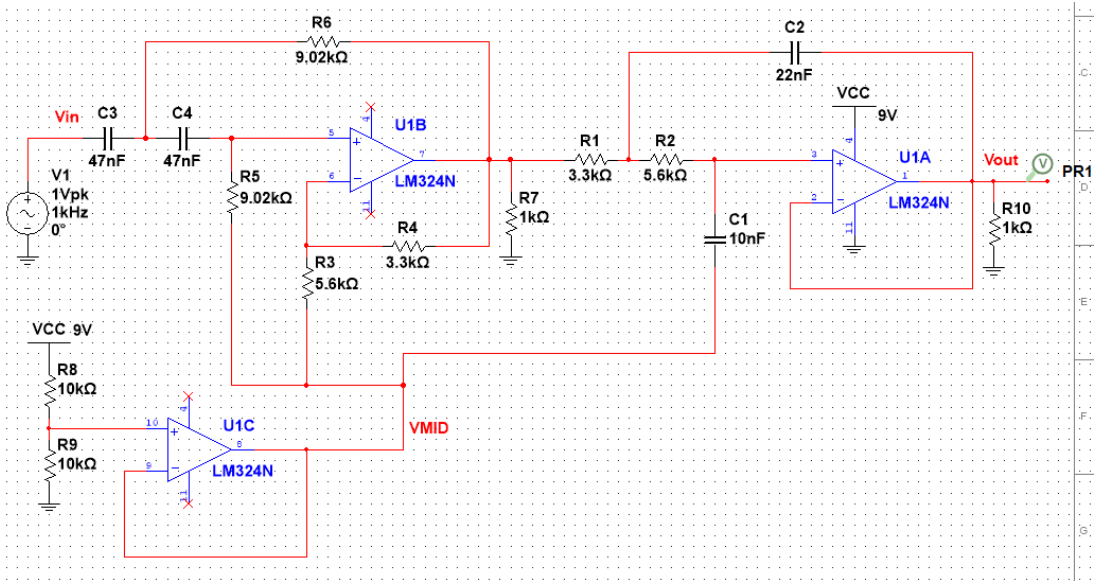


Figure 53: Complete Band Pass Filter

This is now a 4th-order overall band-pass filter formed by cascading a 2nd-order high-pass and a 2nd-order low-pass stage.

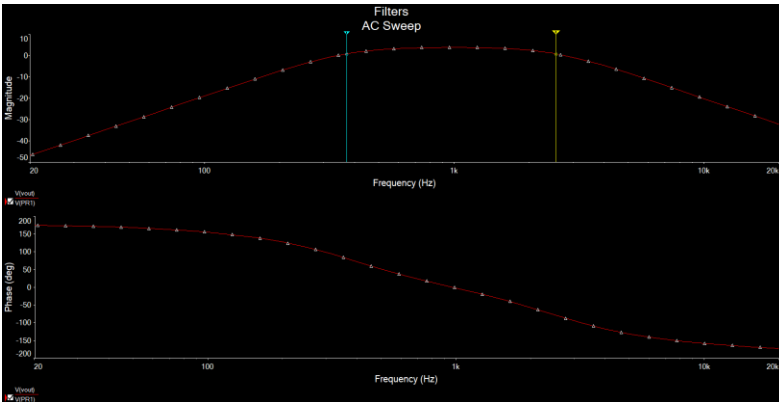


Figure 54: Simulated bode plot for the 4th order Band Pass filter – the cutoff frequencies match almost exactly (Peak magnitude is 3.8568dB, therefore cutoff measured at 856.8mHz)

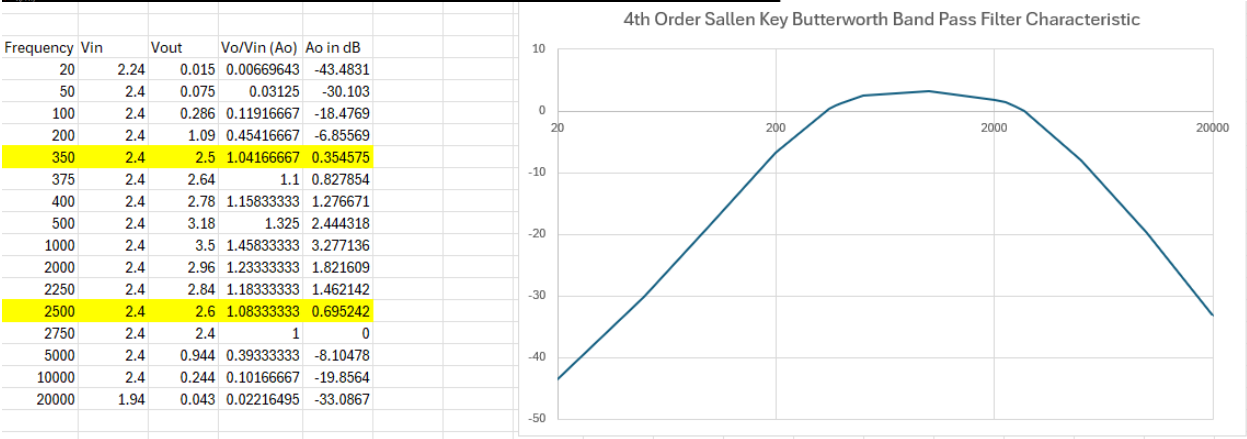
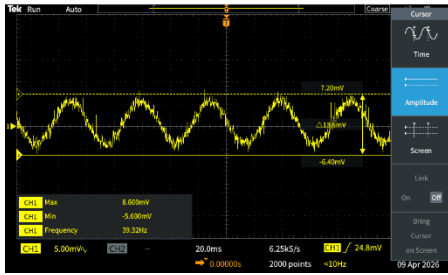
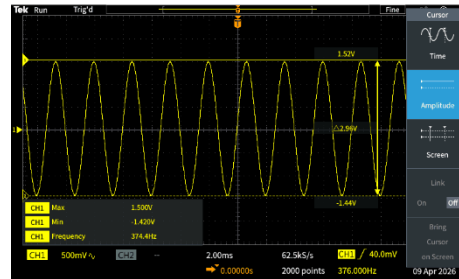


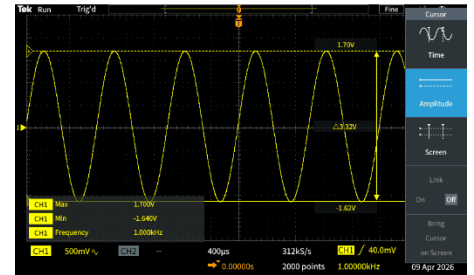
Figure 55: Bode plot for the built band pass filter – note that the lower cutoff has shifted down slightly, however, since it is not a major fluctuation, which can be caused by capacitor tolerances, it can be ignored.



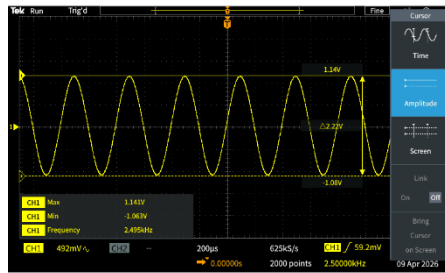
TBS1052C - 18:02:03 08/04/2026



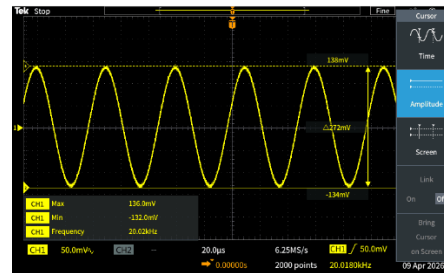
TBS1052C - 18:03:01 08/04/2026



TBS1052C - 18:04:06 08/04/2026

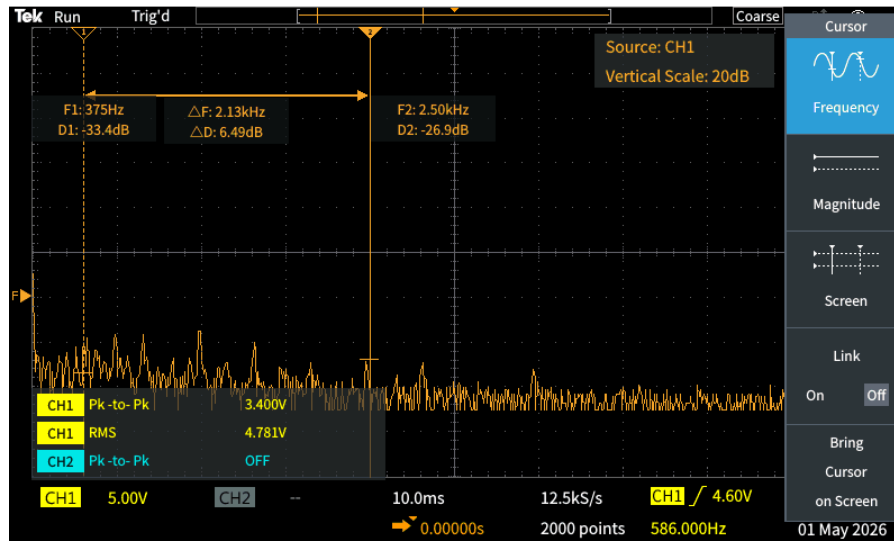


TBS1052C - 18:04:59 08/04/2026



TBS1052C - 18:06:07 08/04/2026

Figures 56, 57, 58, 59 and 60: Filtered waveforms through the BPF at 20Hz, 375Hz, 1kHz, 2.5kHz and 20kHz



TBS1052C - 22:36:24 30/04/2026

Figure 61: FFT of the filtered waveform. As observed here, the prominent frequency peaks are within the 375Hz and 20kHz boundary, which means that the filter functions properly.

High-Pass Filter – Mohammed Alabduljader

1) Filter Selection and Design Objective

It should be noted that the function of the high-pass component in the sound to light conversion system consists of separating the higher parts of the audio spectrum, allowing the higher frequencies to be filtered independently of other lower and band-passed ranges. Considering the general scheme of the sound to light system, it becomes evident that an audio signal at the entry level is first placed into the buffer memory and after that, it is split into three branches representing various frequency ranges. The starting point of the high-pass branch thus needs to follow the end point of the band-pass branch. In the current design, the cut-off frequency is 2.5 kHz.

First of all, a first-order high-pass filter has been considered due to its simple configuration. However, the problem with the one-pole configuration is that its slope for attenuation between low and high frequencies is +20 dB per decade, and thus the differentiation between adjacent bands would be slow. For achieving the goal of this project, we need to clearly distinguish the adjacent frequency bands after their conversion into DC voltage and then driving the LEDs by the corresponding voltages. This is why a second-order filter would provide better results.

In particular, the slope of attenuation in this case will be equal to 40 dB per decade, and also a possibility of controlling the transfer function via a quality factor will emerge. Thus, the second-order Sallen-Key high-pass filter with a Butterworth response was used (Sallen and Key, 1955; Butterworth, 1930).

2) Single Supply Biasing Requirements

In contrast to the dual power supply configuration in an op-amp circuit, the current project employs a single power supply voltage of 9 volts alongside the LM324 op-amp (Texas Instruments, 2015). The audio input signal can be considered an AC signal with a zero volts level, while the op-amp requires operation within the existing 0V to 9V levels.

Consequently, the filtering circuit cannot be grounded in the conventional manner as a dual power supply active filter (Mancini, 2003).

A virtual ground, denoted V_{mid} , was created at half the supply voltage:

$$V_{mid} = \frac{9}{2} = 4.5 \text{ V}$$

The bias voltage has been created by utilizing two identical resistors wired up as a voltage divider from the 9V source. This voltage is made stable with the help of decoupling capacitors. Thus, the AC voltage signal can swing both ways from the bias voltage of 4.5V

and stay in the operating range of LM324. In AC circuit analysis, we take Vmid as the reference point for small signals, hence, instead of connecting our resistors to ground, they will connect to Vmid.

3) First-Order Design Consideration

For a first-order high-pass filter, the cut-off frequency is given by:

$$f_c = \frac{1}{2\pi RC}$$

To get f_c close to 2.5 kHz using capacitors we can find conveniently, we choose $C = 10 \text{ nF}$, and R becomes:

$$R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi(2500)(10 \times 10^{-9})} \approx 6.37 \text{ k}\Omega$$

This proves that it is easily possible to design a system at a frequency range of 2.5 kHz. Nevertheless, while the first order analysis is quite helpful as an initial step, the single pole design could not be used since the rate of attenuation at frequencies above the cut-off would be insufficient to create a proper differentiation with the adjacent band-pass filter.

4) Second-Order Sallen-Key Design

A second-order Sallen-Key high pass filter was selected as the circuit topology for the filter design. The selection was made based on the fact that a second-order transfer function can be clearly realized with a single-stage circuit using one op-amp.

We assume an ideal op-amp with equal-value components: $R_1 = R_2 = R$, $C_1 = C_2 = C$. The high-pass transfer function becomes (Sallen and Key, 1955; Horowitz and Hill, 2015):

$$H(s) = \frac{Ks^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

Where $\omega_0 = \frac{1}{RC}$ and $f_0 = \frac{1}{2\pi RC}$ is the natural frequency of the filter. $Q = \frac{1}{\sqrt{2}} = 0.7071$ for a Butterworth response (Butterworth, 1930; Horowitz and Hill, 2015). And for this Sallen-Key implementation, we achieve it using a gain of $K = 1.586$. The op-amp gain is calculated by $K = 1 + R_3/R_4$ (Mancini, 2003).

5) Component Selection

To keep the design practical for standard laboratory components, the capacitor values were chosen as: $C_1 = C_2 = 10 \text{ nF}$.

Using cut-off frequency of 2.5 kHz, the needed resistors are:

$$R1 = R2 = R = \frac{1}{2\pi RC} = \frac{1}{2\pi(2500)(10 \times 10^{-9})} \approx 6.37 \text{ k}\Omega$$

We can use 6.38 k Ω as 6200 + 180 Ohms, which is available in the lab. This would bring down the cut off frequency slightly:

$$f_0 = \frac{1}{2\pi(6380)(10 \times 10^{-9})} \approx 2.49 \text{ kHz}$$

Which is essentially the same as 2.5 kHz.

To realize Butterworth gain, $K = 1 + R3/R4 = 1.586$. Since $R4 = 10 \text{ k}\Omega$ which is a standard value, we use $R3 = 5.86 \text{ k}\Omega$, which can be divided to 5600 + 270 Ohms, yielding 5.87 k Ω .

$K = 1 + 5.87/10 = 1.587$, which is very close to the Butterworth requirement.

The mid-supply reference was made by two equal 1 k Ω resistors connected between the 9 V rail and the GND, which splits the voltage in two, and creates a node with 4.5 V (V_{mid}).

6) Circuit Operation

The circuit was built on Multisim and is shown in Figure 1 below.

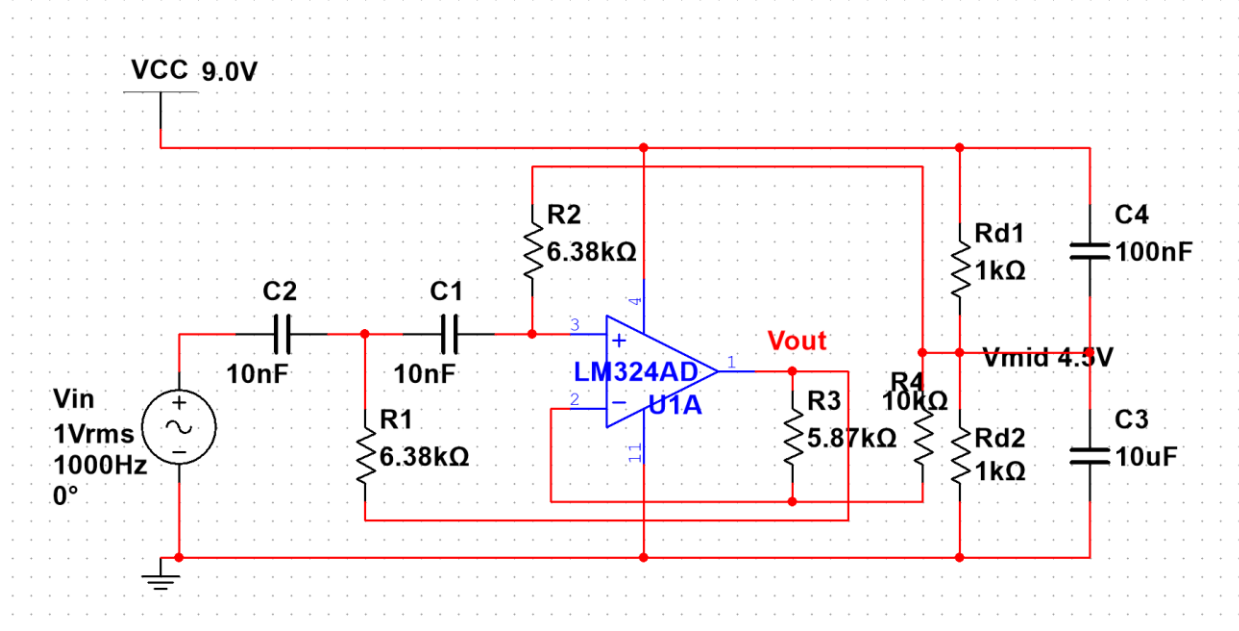


Figure 62: Second Order Sallen-Key HPF Satisfying Butterworth Criteria

In the constructed circuit, the signal travels through two capacitors in a series fashion. As the frequency reduces, the capacitive reactance remains high, and thus much of the signal energy is lost before it reaches the op-amp input. On the other hand, at high frequencies,

the capacitive reactance is low, and consequently, the amount of signal reaching the op-amp input node is higher.

The resistor coupled from the filter node to the output terminal and another resistor from the op-amp input node to Vmid create the second-order filter effect. Here, the non-inverting LM324 stage gives the necessary Butterworth gain. Moreover, the use of the Vmid rather than ground helps to make the entire circuit operate properly in a single power source condition.

This means that, using the circuit, we accomplish two tasks at once: the rejection of signals whose frequencies are below 2.5 kHz, and biasing the signal at 4.5V so that the rectifier stage after can work properly from the available 9V supply voltage.

7) Frequency Response Analysis

Figure 2 below shows the Bode plot and the cursor on the 2.5kHz point.

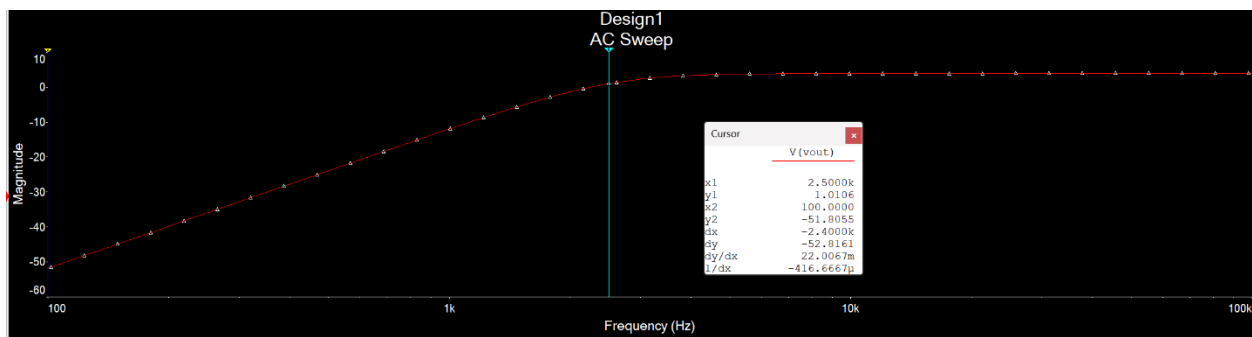


Figure 63: Bode Plot of Circuit with Cursor on Cutoff Frequency

AC sweep analysis validates that the designed circuit operates as a second-order high-pass filter. As seen from the graph, at lower frequencies, the magnitude is highly reduced and gradually increases till it reaches the passband. The slope initially attained at the transition point corresponds to a second-order response, which equals approximately 40 dB/decade.

Since the designed filter does not have a passband gain equal to one, it means that the cut-off frequency will be found by considering the gain of the passband. The passband gain of the designed circuit is:

$$20 \log_{10} 1.587 \approx 4.01 \text{ dB}$$

Therefore, cut-off point corresponds to the magnitude 3 dB below this point: $4.01 - 3 = 1.01$ dB.

From the AC sweep graph, the cursor shows that the magnitude becomes around +1.01 dB exactly at 2.50 kHz. It is in agreement with the design specifications and proves that the values of the resistors and capacitors give the desired crossover point.

Moreover, the phase curve is also consistent with the second-order high-pass circuit's characteristics. The phase begins with nearly $+180^\circ$ at lower frequencies, but as the frequency goes up, the phase changes gradually towards 0° . These features are typical for a second-order high-pass circuit having two zeros at the origin.

The fact that there is no significant peaking in the cut-off region means that the gain is close to the idealized Butterworth value.

8) Theoretical vs Practical Comparison

The filter circuit was constructed on a breadboard employing the materials described in Materials Used, and then tested over the range of audio frequencies using a function generator and an oscilloscope. A sinusoidal input voltage of 3 Vpp was applied to the circuit at each testing frequency and output voltage magnitude noted. Gain in both V/V and dB was subsequently calculated.

Frequency (Hz)	Vin (Vpp)	Vout (Vpp)	Gain (V/V)	Gain (dB)
20	2.92	0.04	0.014	-37.27
50	3.04	0.05	0.016	-35.68
100	3.04	0.05	0.016	-35.68
200	3.04	0.08	0.026	-31.60
500	3.12	0.20	0.064	-23.86
1000	3.12	0.83	0.266	-11.50
2000	3.12	2.50	0.801	-1.92
2250	3.12	2.96	0.949	-0.46
2500	3.12	3.28	1.051	+0.43
2750	3.12	3.50	1.122	+1.00
5000	3.04	4.44	1.461	+3.29
10000	3.04	4.54	1.493	+3.48
20000	2.40	3.62	1.508	+3.57

Table 3: Measured input/output voltages and gains across the test frequency range.

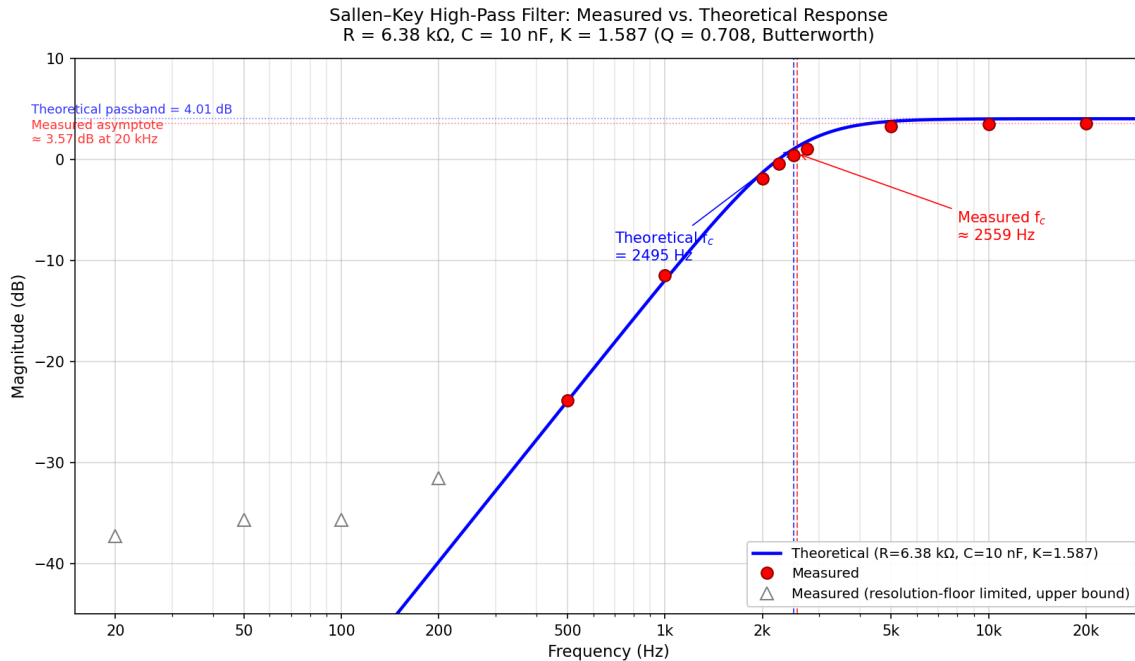


Figure 64: Comparison of the theoretical Butterworth response (blue) with the experimentally measured points (red). The two vertical dashed lines mark the theoretical ($\approx 2495 \text{ Hz}$) and measured ($\approx 2559 \text{ Hz}$) cutoff frequencies. Open triangles denote measurements where the oscilloscope resolution floor ($\approx 50 \text{ mV}$) limits the readable V_{out} — these points are upper bounds on the true filter response.

Cutoff frequency

Using the -3 dB level relative to the experimentally obtained asymptotic passband gain ($\sim 3.57 \text{ dB}$ @ 20 kHz), it can be seen that the experimentally obtained cutoff frequency occurs somewhere between the 2.50 kHz point (gain level = $+0.43 \text{ dB}$) and the 2.75 kHz point (gain level = $+1.00 \text{ dB}$). Using logarithmic interpolation, the experimentally obtained cutoff frequency is thus approximately 2559 Hz . The expected frequency is 2495 Hz , calculated using $f_0 = 1/(2\pi RC)$, which has an error of only $+2.6 \%$ compared to the experiment.

Passband gain

The gain has not reached the asymptotic state as high as 20 kHz (where gain has increased from 3.29 dB at 5 kHz to 3.48 dB at 10 kHz to 3.57 dB at 20 kHz), but slightly higher than that. Using 3.57 dB as the minimum gain, the measurement of K is 1.508 V/V , while theoretically, K is supposed to be 1.587 V/V (or 4.01 dB), a difference of 0.4 dB . It appears likely that the discrepancy could be due to two factors: (i) limited gain bandwidth ($\approx 1 \text{ MHz}$) of LM324, which causes a fall in gain even at the upper audio frequencies; and (ii) the $\pm 5 \%$ tolerance in the value of R_3 , whose low value will result in low gain

Roll-off slope and the low-frequency floor

Between the transition band (500 Hz to 2 kHz) and below it, the observed slope equals about 36 dB per decade, matching well the theoretical value of -40 dB per decade for a second-

order high-pass filter. However, below 500 Hz, the slope appears to be around 12 dB per decade, which is an artifact of the measurements. The recorded values of the V_{out} voltages at 20 Hz, 50 Hz, 100 Hz, and 200 Hz lie very close to the vertical resolution limit of the oscilloscope ($\approx 40\text{--}80$ mV). This means the recorded values are the maximum possible values of the voltages and are the upper bound on the filter output voltage. Thus, the attenuation is much higher than that calculated. These four data points are marked by empty triangles in Figure 3.

Summary

The measured response of the filter compares favorably with its theoretical design as a Butterworth filter, considering the limitations of the test bench equipment: the cut-off frequency matches that of the design very closely, lying within $\pm 2.6\%$ error from the ideal value of 2.5 kHz; the transition region matches the theoretical approximation of 40 dB/decade slope, while the pass-band behaves as expected, exhibiting flat behavior without peaking effect and a small gain deficit of 0.4 dB due to limited gain-bandwidth product of op-amp and resistor tolerances. The observed flattening of the slope in the stop-band below 500 Hz is an instrumentation resolution issue and not a flaw in the circuitry.

Bill of Materials

Component	Quantity	Cost
Resistors (various values)	54	
Capacitors (various values)	16	
LM324N IC	6	
LM3914 IC	3	£ 6.00
BGYR LED bar graph	3	£ 6.00
100 k Ω potentiometer	1	
10 k Ω potentiometer	3	
Connector wires	-	
Breadboard	1	
3.5 mm TRS audio jack	1	£ 7.00
Total		£ 19.00

Table 4: Bill of materials for the project

A total of 54 resistors were used, distributed as follows: one 270 Ω , three 560 Ω , one 680 Ω , five 1 k Ω , four 1.2 k Ω , four 1.5 k Ω , five 2.2 k Ω , one 3.3 k Ω , three 5.6 k Ω , two 6.8 k Ω , four 8.2 k Ω , nineteen 10 k Ω , one 22 k Ω , one 100 k Ω , and one 220 k Ω . In addition, 16 capacitors were used, comprising four 1 μF , one 3.3 μF , one 10 μF , four 47 nF, three 33 nF, two 10 nF, and one 100 nF capacitors.

Active Peak Detector

After each filter stage, an active peak detector was implemented. A normal peak detector which was initially suggested in the block diagram given in the project brief was determined to be not suitable for this project because a voltage drop of around 0.6V is usually detected across the diode, and since the later implemented LED Driver stage is sensitive to levels of voltage the diode drop needs to be mitigated. An active peak detector, which uses a feedback path between two op-amps, does not contain this diode drop and is ideal to be used for this project.

The capacitor and resistor values were selected based on the estimated satisfactory smoothing behaviour of the filtered signals. The RC time constant is given by:

$$\tau = RC = 100k\Omega \times 1\mu F = 0.1s$$

This corresponds to a discharge time of approximately 100ms, which is suitable for audio signals. It is long enough to smooth out the high frequency audio waveforms while still allowing the output to follow slower changes in amplitude.

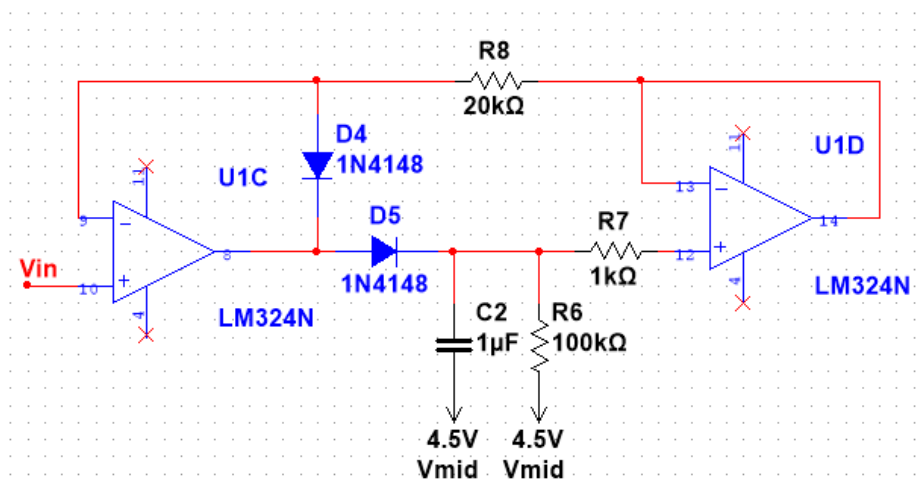


Figure 65: The implemented active peak detector. The filtered signal comes in through V_{in} , where the capacitor track the amplitude peaks of the signal. The capacitor and the 100k bleed resistor are referenced to V_{mid} .

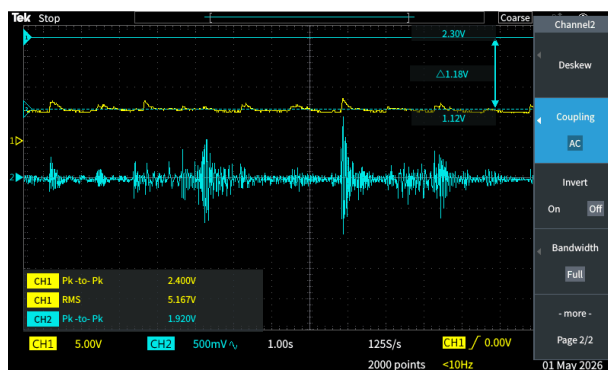


Figure 66: Filtered signal (blue waveform) and the output of the peak detector (yellow waveform)

Gain Stages

A variable gain stage is required to be implemented for each filter because the level of gain already existing with each filter can fluctuate based on component tolerances and breadboard parasitics. A variable 'master gain' stage is also implemented at the input stage of the circuit because some sources of audio may be softer in volume/amplitude than others, which may cause the LED bar graphs to not illuminate to their maximum potential.

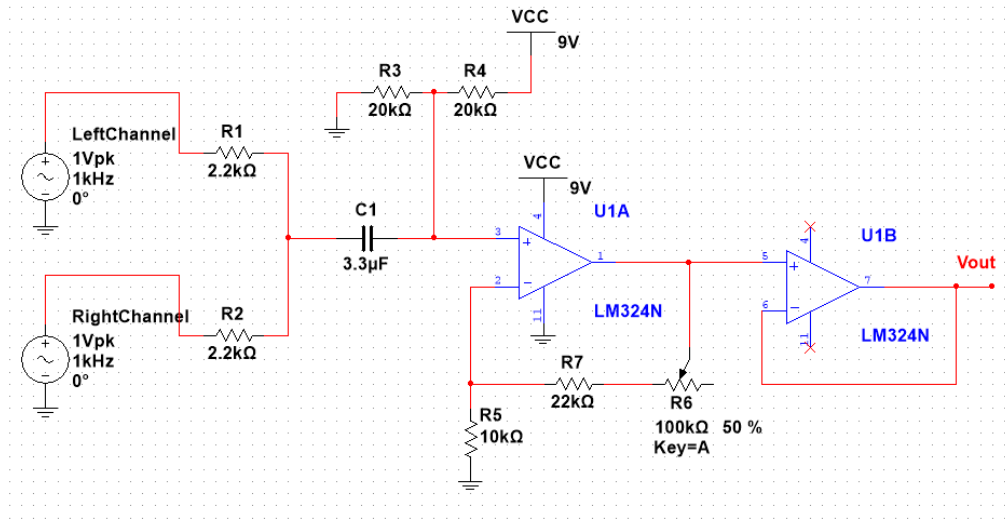


Figure 67: Input stage with the master gain

Here, a 100k Potentiometer is added in series with a 22k resistor in the non-inverting op-amp before the voltage follower. Therefore, the master gain can go from:

$$A_{o,min} = 1 + \frac{R_{2,min}}{R_1} = 1 + \frac{22}{10} = 3.2 \quad \text{up to:} \quad A_{o,max} = 1 + \frac{R_{2,max}}{R_1} = 1 + \frac{122}{10} = 13.2$$

Next, individual gain stages for each of the filters after the active peak detector stage was implemented. The observed maximum swings of the output of the active peak detector was around 1V to 2V from Vmid, and the op-amp started showing clipping behaviour when the signal was around 4V from Vmid. Therefore, the individual gain stages were configured so that the signal can be amplified up to around 2.7V from Vmid, which will be the maximum signal limit that goes into the LED Driver IC mentioned later.

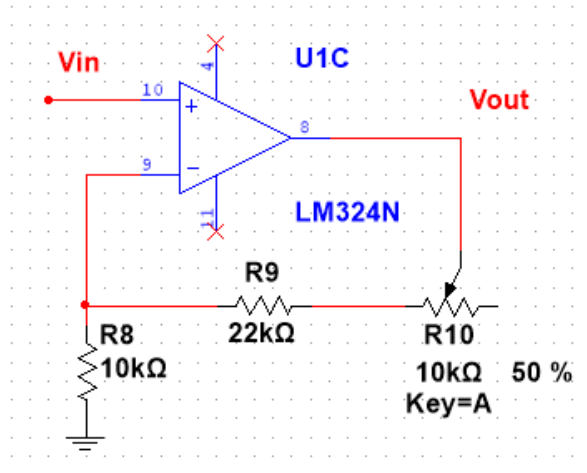


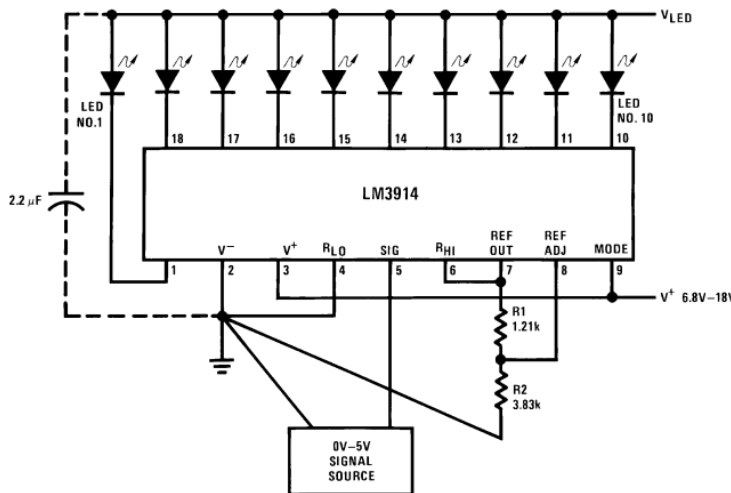
Figure 68: Simple Individual gain stage circuit. Note that in the actual implementation some resistor values were switched out based on observed functionality

Here, the gain can go from:

$$A_{o,min} = 1 + \frac{R_{2,min}}{R_1} = 1 + \frac{22}{10} = 3.2 \quad \text{up to:} \quad A_{o,max} = 1 + \frac{R_{2,max}}{R_1} = 1 + \frac{32}{10} = 4.2$$

LED Driver IC

Research was conducted into what IC can be used to drive the LED bar graphs in the music visualiser arrangement as required. The LM3914 Dot/Bar Display Driver was selected, because it allowed a bar graph meter arrangement which also regulates the current flowing into the LEDs, which eliminates the need for extra resistors to be added with the LED bar graphs. The bar graph meter circuit provided in the datasheet was taken as a reference to



build our own arrangement.

Figure 69: LM3914 Bar Graph Meter arrangement provided in the datasheet

From the given equation and the maximum signal value of 2.7V from Vmid (around 7.2V), we can determine the values for R1 and R2.

$$2.7 = 1.25 \left(1 + \frac{R_2}{R_1} \right)$$

$$\text{Ref Out } V = 1.25 \left(1 + \frac{R_2}{R_1} \right)$$

$$I_{LED} \approx \frac{12.5}{R_1}$$

Note: Grounding method is typical of all uses. The 2.2μF tantalum or 10 μF aluminum electrolytic capacitor is needed if leads to the LED supply are 6" or longer.

$R1 = 1.2k$ and $R2 = 1.5k$ selected, which puts the range that the LM3914 sees to turn on the LEDs around 2.812V, therefore the range becomes 4.5V to around 7.3V.

The current flowing into the LEDs can be calculated using the given equation as well:

$$I_{LED} = \frac{12.5}{1200} \approx 10.4mA$$

Therefore LED brightness would generally remain normal and safe.

Since the LM3914 is powered by the V- and V+ pins, V- is kept at true ground and V+ at 9V. The RLO pin, which is the pin that determines the lower end of the voltage range that the IC sees, is set to Vmid. The remaining connections are made exactly the same as given.

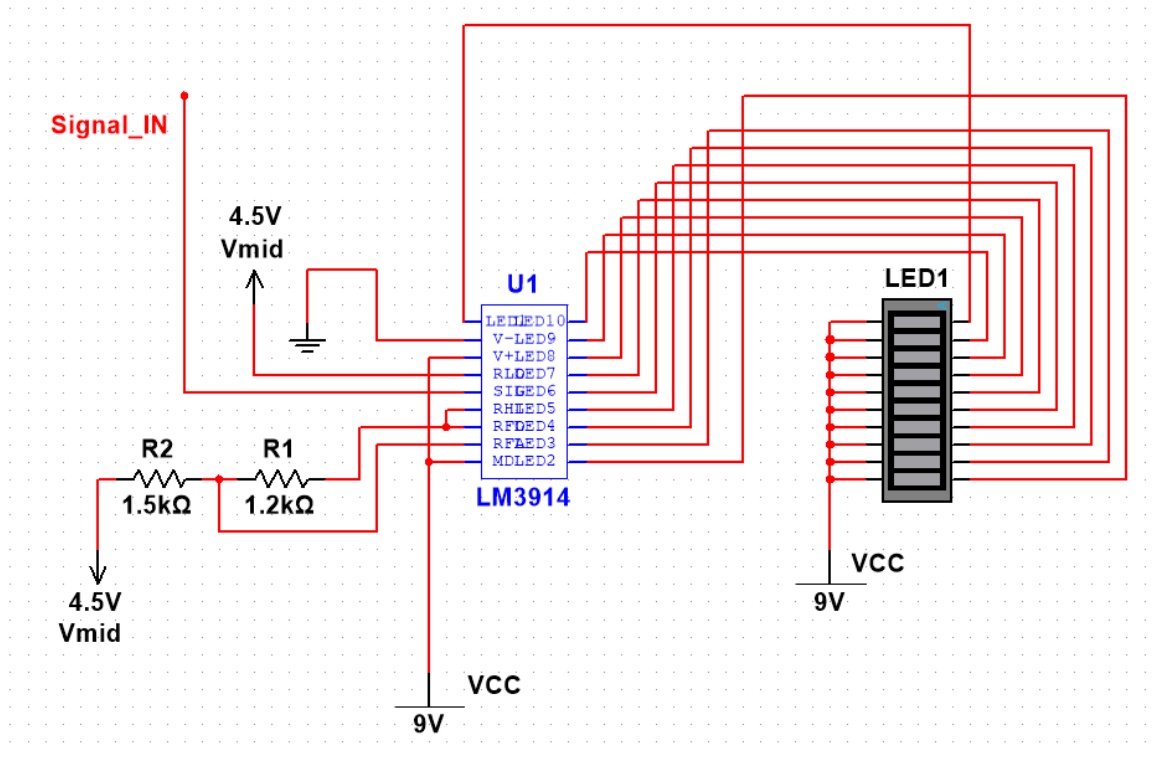


Figure 70: LM3914 wiring using a custom multisim component

Full Circuit Implementation and Function

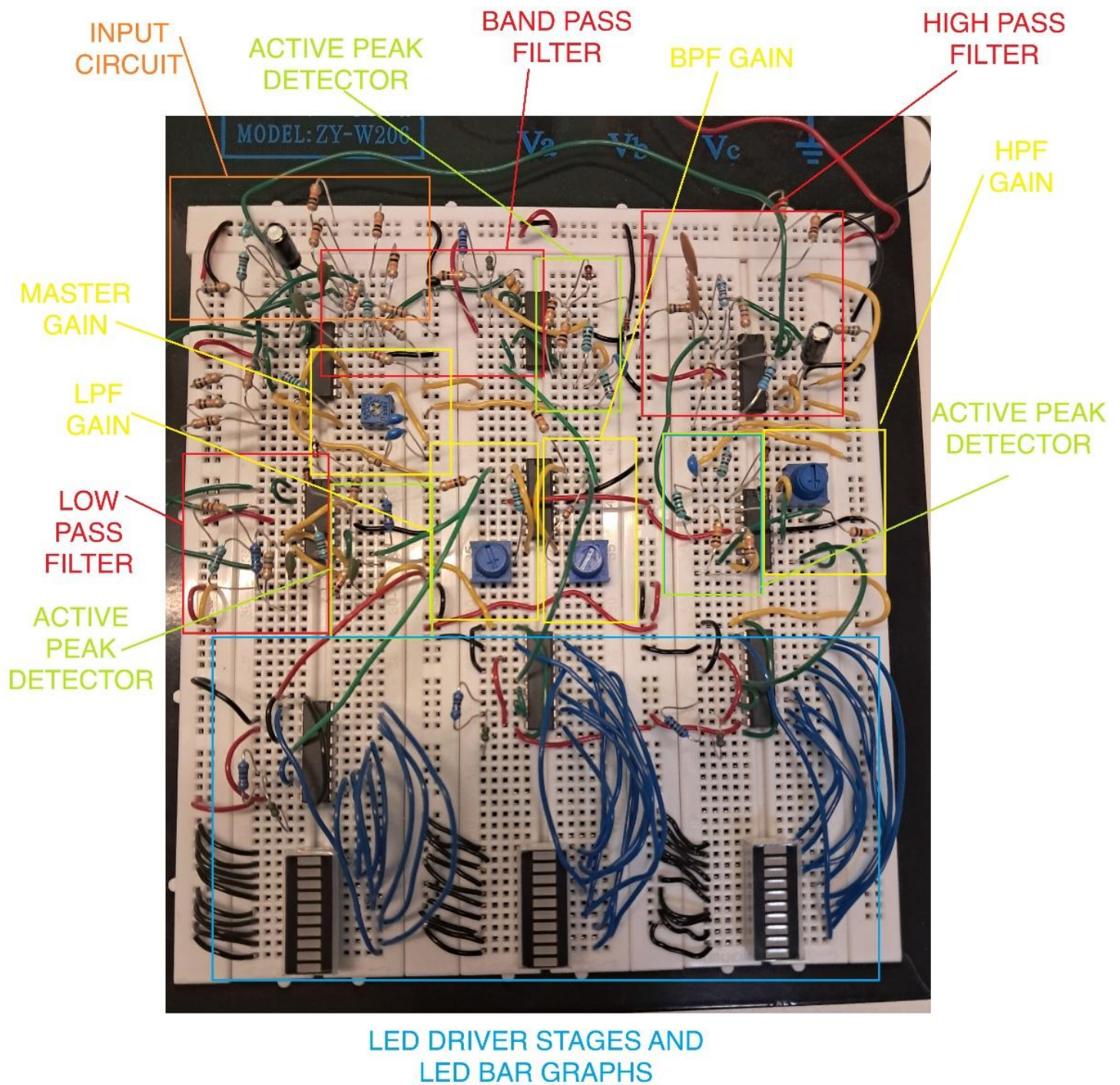


Figure 71: The complete assembled circuit with marked sections

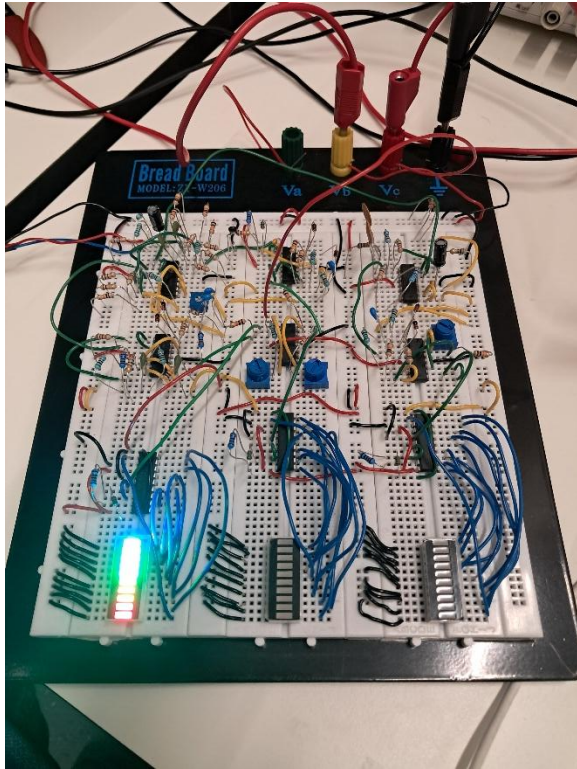


Figure 72: LED Behaviour at 20Hz

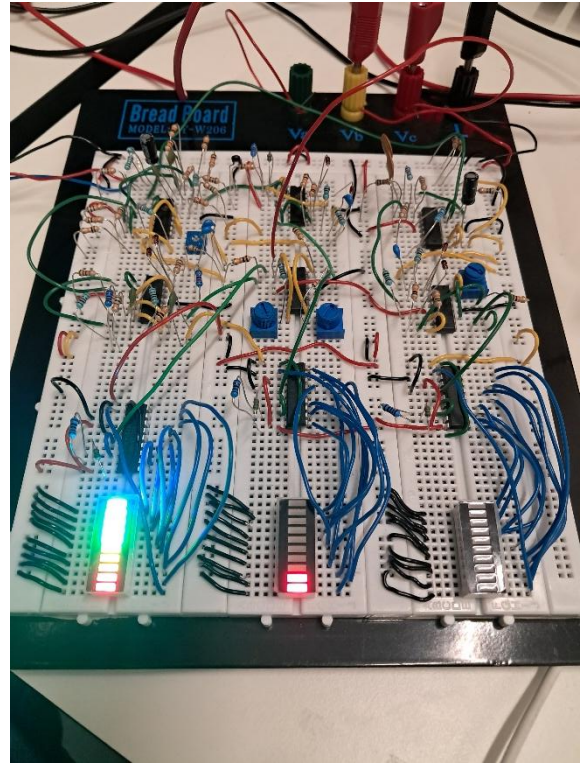


Figure 73: LED Behaviour at 175Hz

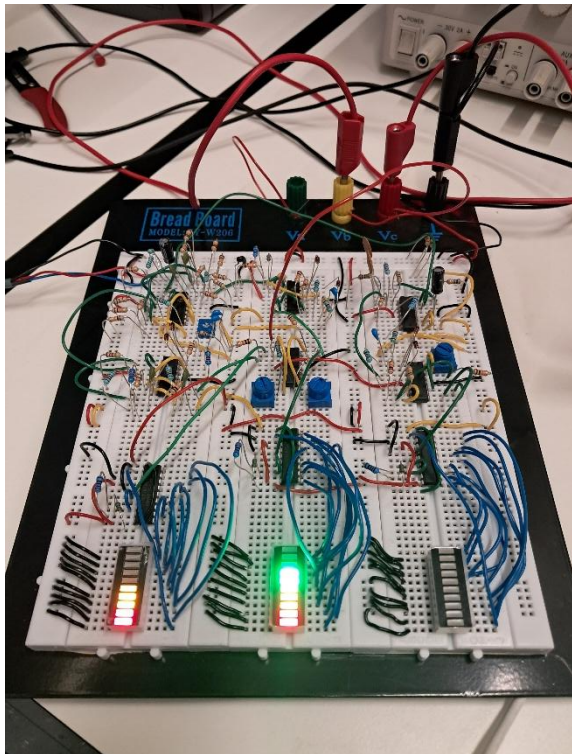


Figure 74: LED Behaviour at 375Hz

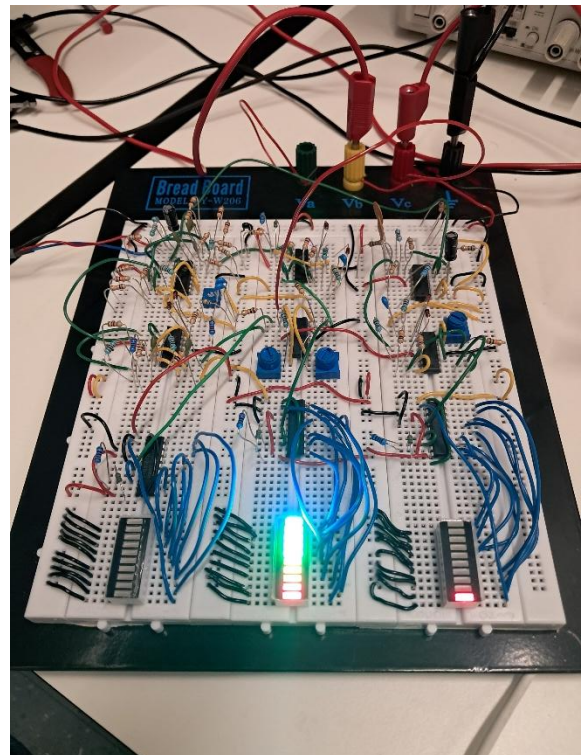


Figure 75: LED Behaviour at 1kHz

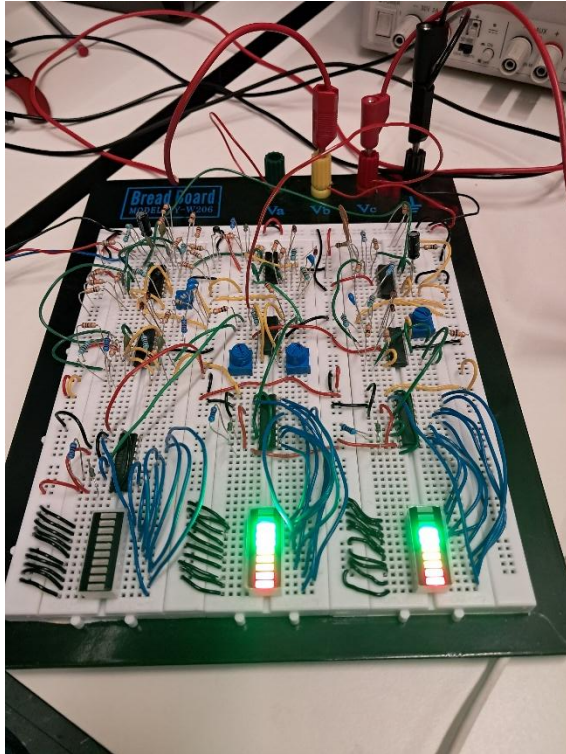


Figure 76: LED Behaviour at 2.5kHz

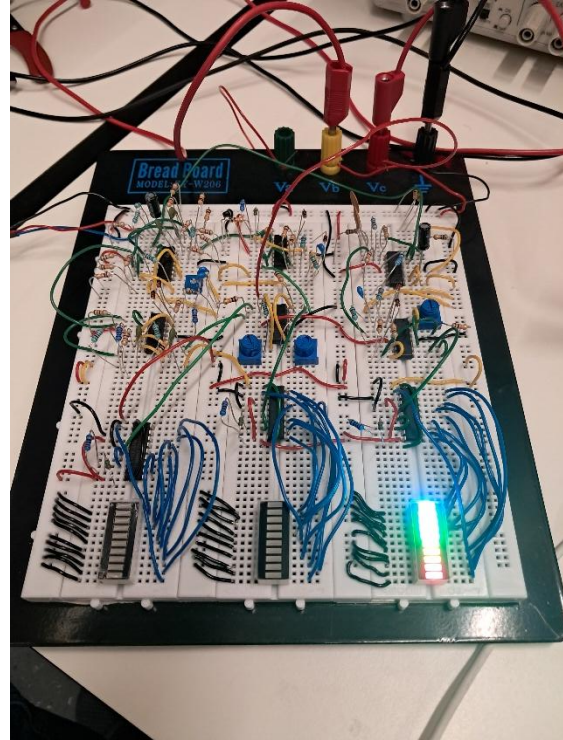


Figure 77: LED Behaviour at 10kHz

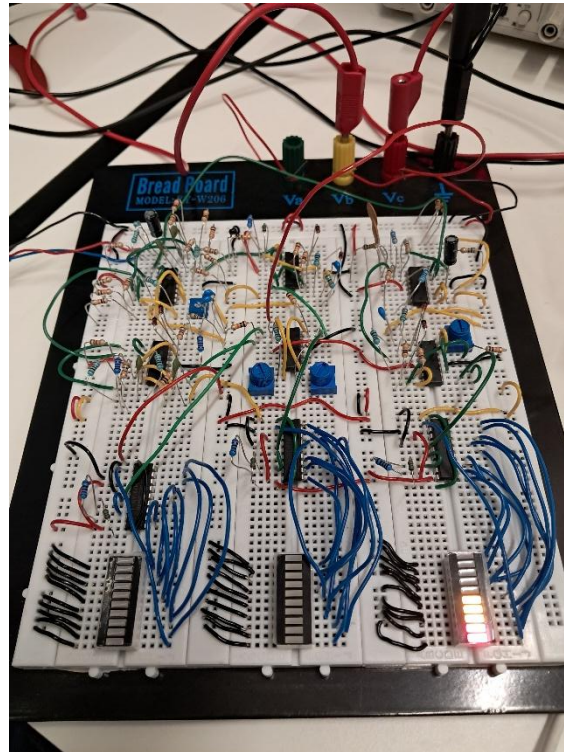


Figure 78: LED Behaviour at 20kHz

Discussion

This sound-to-light 3 band audio visualiser system consists of three filters as low-pass, band-pass and high-pass, each followed by signal conditioning stages (gain, peak detectors etc) and an LM3914-based LED bar graph implementation. An input audio signal through an audio jack would be split into the mentioned frequency bands and their relative intensities would be visualized in real time.

During development, several design choices were revised. Initially it was proposed to implement an absolute value circuit right before the active peak detector stage to improve the 'smoothing' function of the peak detector and reduce the potential flickering of the LED stages, but it was deemed to overcomplicated the project, and after many iterations and testing, it was abandoned.

Configuring the signal going into the LM3914 LED driver was also a key design stage. At first, we attempted to use a difference amplifier to remove the V_{mid} bias from the signal and make it centered around 0V so that we can add a lot of gain to the signal, thus making the resultant LED behaviour more pronounced. However, the difference amplifier failed to cleanly remove the 4.5V bias, leaving around 100mV of an offset, therefore, this approach was abandoned, and it was decided to use 4.5V as the reference voltage for the LM3914 (Pin 2).

The Active Peak Detector stage also involved some design choices. A balance had to be achieved between responsiveness and stability of the LED output. Choosing a smaller RC time constant would have resulted in heavily flickering LEDs, and larger values would have produced excessive lag. Through experimentation, we selected suitable resistor-capacitor combinations (1uF-2uF and 100k) to achieve a pleasing visual output.

The final system performed very satisfactorily under real audio input conditions. Each band responded distinctly to its corresponding frequency range, and the LED outputs provided a clear visual representation of signal amplitude.

A Summary of What Went Wrong and How It Was Fixed

- Cutoff Shifts

Erroneous cutoffs or shifted cutoffs were observed when the built circuit capacitors etc were incorrectly referenced to true ground rather than V_{mid} , and with capacitor tolerances, loading from the crossover distortion mitigating resistors and breadboard parasitics. This was solved through proper component referencing, adjusting calculated component values to counterbalance the cutoff shift, increasing the resistance used for crossover distortion and cleaner wiring.

- Diode Voltage Drop in Peak Detection

Initially, a normal peak detector was implemented, which included a 0.6V voltage drop, which reduced signal amplitude and consequently LED resolution. This was resolved by adding different gain stages throughout the circuit as mentioned in previous sections.

- Gain Stage Instabilities

The gain stage produced incorrect DC outputs (7V constant output etc) due to improper biasing and loading effects. This was solved by correcting input referencing and ensuring proper grounding.

- LM3914 input loading

Connecting the signal directly to the LM3914 caused unexpected voltage shifts, which was resolved primarily by adding a voltage follower before every signal output stage.

Reflections by Each Group Member

Yasandu Kalansooriyage

The band-pass stage required the most effort to stabilise because of the interaction between the second order low and high pass filters that are cascaded together. They were individually tested first, verified for errors and was confirmed to be working. However, after integrating it with the absolute value circuit and LED driver there were multiple issues related to DC offset and signal scaling etc.

The effect of the most insignificant voltage offset propagating through the system and being amplified by later stages was one of the main challenges faced. Through careful inspection of each nodes through an oscilloscope, diagnosing faults became more intuitive, and I shifted from a trial-and-error method to a very targeted debugging method based on experience gained. By the end of the project I became able to immediately understand almost any issue related to the circuit, guess resistor values at first glance and spot wiring mistakes.

The limitation of theoretical designs when applied to real components was also a very important point that I noted. Due to tolerances, values like cutoff frequencies and gain levels often shifted around, and were required to be adjusted below or beyond calculated values.

In conclusion, working on the band-pass stage and the circuit as a whole significantly improved my personal understanding of signals and systems, and analogue signal processing, and would be a very helpful benefit in learning Digital Signal Processing in the future.

Mohammed Alabduljader

This filter design proved to be a practical lesson in balancing theoretical ideals with real-world constraints. The first-order topology was quickly abandoned because its 20 dB/decade roll-off would not provide sufficient separation between adjacent frequency bands. Moving to a second-order Sallen-Key Butterworth design with a 2.5 kHz cut-off was therefore necessary. A notable challenge was operating from a single 9 V supply; this was resolved by creating a 4.5 V virtual ground, allowing the AC signal to swing correctly. Simulation results aligned well with theory, but breadboard measurements revealed minor discrepancies. The measured cut-off frequency (≈ 2559 Hz) deviated by only +2.6% from the target, and the 0.4 dB passband gain deficit could be reasonably attributed to the

LM324's limited gain-bandwidth product and resistor tolerances. The apparent flattening of the roll-off below 500 Hz was correctly identified as an oscilloscope resolution artefact rather than a circuit flaw. Finally, integrating the filter into the full signal chain—variable gain, precision rectification, and LED driving—reinforced the importance of considering the entire system, not just individual stages. Overall, this project demonstrated that successful analog design relies as much on careful measurement and component awareness as on theoretical calculation

Conclusion

A three-band sound-to-light audio visualiser was developed using analogue filter stages, signal conditioning circuits, and LM3914 LED drivers. The system receives an audio signal from a line-out source, separates it into low, mid, and high frequency bands, and displays their relative amplitudes on LED bar graphs in real time.

During operation, the signal range was approximately 4.5 V to 7.3 V, which matches the LM3914 input range referenced to V_{mid} . The peak detection stage employed a time constant of approximately 100 ms, balancing responsiveness and stability. This configuration enabled the LEDs to respond smoothly to signal changes without noticeable flicker.

The measured results were generally close to the expected values, although some differences in cutoff frequency and gain were observed. These were mainly due to component tolerances, breadboard effects, and the limited performance of the LM324 at higher frequencies. During development, issues such as crossover distortion and bias instability appeared, but these were resolved through iterative testing and modification of the circuit.

The system worked reliably overall, but some limitations became clear. The LM324 introduced non-ideal behaviour, particularly at higher frequencies, and small variations in component values affected the accuracy of the filters. A more refined version of the design would benefit from being implemented on a PCB, with improved grounding and layout, and possibly using higher-performance op-amps.

In closing, the project demonstrated how filter design translates into practical circuit implementation while also showing the importance of testing and debugging based on real-world behaviour, and was an excellent learning experience for the subject of Signals and Systems.

References

- Butterworth, S. (1930) 'On the theory of filter amplifiers', *Experimental Wireless and the Wireless Engineer*, 7, pp. 536–541.
- Horowitz, P. and Hill, W. (2015) *The Art of Electronics*. 3rd edn. Cambridge: Cambridge University Press.
- Mancini, R. (ed.) (2002) *Op Amps for Everyone: Design Reference*. 2nd edn. Burlington, MA: Newnes.
- Sallen, R.P. and Key, E.L. (1955) 'A practical method of designing RC active filters', *IRE Transactions on Circuit Theory*, 2(1), pp. 74–85.
- Texas Instruments (2015) LM324 Quadruple Operational Amplifier datasheet (SLOS066W). Available at: <https://www.ti.com/lit/ds/symlink/lm324.pdf> (Accessed: 21 April 2026).
- Texas Instruments (2013) *LM3914 Dot/Bar Display Driver datasheet (Rev. B)*. Available at: <https://www.ti.com/lit/ds/symlink/lm3914.pdf> (Accessed: 2 April 2026).
- Mancini, R. (2002) *Analysis of the Sallen-Key Architecture*. Application Report SLOA024B. Dallas, TX: Texas Instruments. Available at: <https://www.ti.com/lit/an/sloa024b/sloa024b.pdf> (Accessed: 28 April 2026).